

An Integrated World Model for Strategic Planning and Scenario Analysis

Nazar S. Sotiriadi^{a,*}, Andrei A. Osiptsov^{a,b}

^a*JSC Sberbank of Russia, Kutuzovsky avenue, 32, Moscow 121170, Russian Federation*

^b*Skolkovo Institute of Science and Technology (Skoltech), Bolshoy Boulevard 30, bld. 1, Moscow 121205, Russian Federation*

ABSTRACT

This paper describes an integrated simulation model of the world, in which the economy, climate and carbon cycle, warming damages, natural resources, society and politics, interstate relations, culture, crisis and finance mechanisms are coupled within a single computational loop. The model is organized as a system of interacting countries and includes 57 country agents plus several regional aggregates. Simulation advances year by year: each block updates its state annually, and the results are passed to the next year and to adjacent blocks. The aim is not to deliver a point forecast of the future but to provide a tool for strategic planning and scenario analysis: playing out coherent development scenarios, comparing alternative policy decisions, quantifying uncertainty through ensembles, and detecting weak signals of approaching structural shifts – including for the management of systemic risks. We set out the problem and the model's place among existing approaches, its architecture and the mathematics and empirical basis underlying each block, the calibration and validation procedure with the resulting metrics, sensitivity analysis and ensemble uncertainty quantification, an explicit list of limitations, and typical use cases, including a decision-game layer built on the same core whose automated playthrough distributions double as a stress test of the coupled dynamics. The economic and climate blocks are calibrated on 1990–2023 historical data and pass retrospective validation. The society, politics, and geopolitics blocks are assessed against a more modest criterion – improvement over consensus expert forecasts.

Keywords: integrated assessment model (IAM), strategic planning, scenario analysis, uncertainty quantification, risk management, weak signals.

1. Questions the Reviewer may ask

1. How does history matching discard parameter regions? What observations are used as constraints? How many simulations per parameter set?
2. What is the computational cost? The paper mentions 500-member ensembles and 33-parameter screening. How long does this take? What hardware?
3. Why 57 countries? The choice of 57 countries plus regional aggregates is explained as "the 50 largest economies + residuals + Taiwan." But why exactly 50? Is this a threshold for coverage or a convenience?
4. How is "social tension" defined operationally? It's a latent variable. How is it operationalized to be comparable across countries?
5. What about China, Russia, Iran unique political economies? The model treats all countries with the same functional forms, but these have state-owned enterprises, capital controls, and unique governance structure might require different parameters.
6. How does the model handle international spillovers? The paper mentions "bilateral trade follows a gravity law" but doesn't specify if trade imbalances are tracked.
7. How are submodels for finance, technology diffusion and pandemic are specified?

✉ sotiriadi.n.s@gmail.com (N. S. Sotiriadi)
ORCID(s):

[Addressed: (1) *history matching* — Section 4 (calibration principles, expanded); (2) *computational cost* — new paragraph at the end of Section 5; (3) *choice of 57 actors* — Section 5 (ensemble paragraph) and footnote in Section 2.1; (4) *operational definition of social tension* — Section 2.6; (5) *heterogeneous political economies* — Section 2.6 and Limitations; (6) *spillovers and trade imbalances* — Section 2.8 (block interactions); (7) *finance / technology diffusion / demography-and-pandemic* — Sections 2.2 and 2.7.]

2. Introduction

A more detailed comparison table with specific features, not just checkmarks. 1. The World3 model (Meadows et al.) had system dynamics of global aggregates. 2. The T21 model (Millennium Institute) has country-level detail. 3. The E3ME model (Cambridge Econometrics) couples energy, economy, and environment. What exactly is new? The LLM integration? The crisis module? The specific coupling?

[Addressed: Table 1 is now a feature table (paradigm, resolution, coverage, gaps) rather than a checkmark matrix; World3, T21, and E3ME are included, and the closing paragraph of the Introduction states explicitly what is new.]

Strategic planning and scenario analysis over long horizons — from government policy to corporate strategy — require a tool, in which climatic, economic, and socio-political processes are considered jointly rather than in isolation; the same need arises in the management of systemic risks. In practice, an analyst must rely on a patchwork of disparate models: climate models describe atmospheric physics but contain almost no finance or politics; integrated assessment models (for example, the DICE, RICE, and FUND families) couple the economy and climate but typically lack sovereign finance, interstate relations, and social dynamics [Nordhaus, 2017]; macroeconomic models describe the monetary sphere in detail but ignore natural capital. As a result, the mutual influence of these domains — for example, the path from warming to falling crop yields, rising prices, social tension, and on to political instability and conflict — falls outside the scope of any single model.

Yet it is precisely the coupling of these channels that determines how long-run development scenarios unfold and where systemic risks accumulate. The absence of a coherent model spanning climate, the economy, society, and geopolitics at once is a substantive gap, felt in strategic planning and risk management alike, where the task is not to assess an isolated shock but to trace its propagation across interconnected subsystems and to compare alternative development trajectories. This paper describes a model built to fill that gap. We emphasize a principled stance from the outset: the model does not claim to be an infallible predictor. Assessments of multi-year forecasts show that even professional growth forecasts at horizons of two to five years often fail to beat a naive extrapolation of the mean [Celasun et al., 2021]. *A realistic goal therefore is not in point forecast but in the ability to reproduce scenarios coherently, with proper tracking of the measure of uncertainty along each scenario.*

The novelty of this work lies not in any single block but in the integration and the breadth of coupling; stated exactly: (i) a single annual loop that carries the state through the economy, climate, resources, society, politics, and interstate relations *including discrete crises*, where every listed comparator stops at two or three of these domains; (ii) uncertainty quantification (ensembles, history matching) built into the workflow rather than added post hoc; and (iii) an optional strategy layer in which large-language-model agents set country policies while all consequences are computed by the deterministic core. The crisis module and the language-model layer are components of this integration, not the novelty claim by themselves. Integrated assessment models (IAM) such as DICE and FUND and large modeling frameworks (GCAM, MESSAGEix, REMIND) couple the economy and climate in detail but contain no sovereign finance, interstate relations, or socio-political dynamics; the classic world-system models (World3) lack an explicit price economy and a modern-grade climate. The distinguishing feature of the proposed model — the Global Integrated Model (GIM) — is that it carries the loop through to society, geopolitics, and discrete crises, and embeds large-language-model agents that play as countries (Table 1, Section 8); a controlled benchmark that quantifies this advantage against sectoral models is given in Appendix B.

The coverage comparison (Table 1) is usefully complemented by a comparison along *methodological axes*: the listed models address different problems, so a direct “better–worse” ranking is inappropriate, and the distinction is better drawn by solution paradigm, treatment of uncertainty, and object of validation. In terms of *solution paradigm*, the DICE / RICE / FUND family and REMIND rest on welfare optimization

Table 1

Feature comparison with representative integrated, energy-economy, and world-system models. Each cell states the specific capability rather than a checkmark; the last column lists the domains that fall outside each model's scope but inside the scope of the question this paper addresses (shock propagation across economy, climate, society, and geopolitics).

Model	Paradigm	Domains covered	Spatial units	Not represented
DICE / RICE	welfare optimisation	climate + aggregate economy, quadratic damages	world (RICE: 12 regions)	prices, finance, resources, society, politics, conflict, discrete crises
GCAM	recursive-dynamic market equilibrium	energy–land–water–climate detail	32 regions	sovereign finance, social and political dynamics, interstate conflict
MESSAGEix	energy-system optimisation (+MACRO)	energy technology detail, land use, climate	11+ regions	society, politics, geopolitics; crises only as scenario inputs
E3ME	econometric simulation [Mercure et al., 2018]	energy–economy–environment, demand-led macro	~70 regions	interstate relations, conflict, endogenous discrete crises
T21 / iSDG	system dynamics [Millennium Institute, 2002]	country-level economy–society–environment (SDG scope)	one country per model	interstate trade network, geopolitics, multi-country shock propagation
World3	system dynamics [Meadows et al., 1972]	global stocks: population, capital, resources, pollution	global aggregates	price economy, countries, modern climate module, politics
GIM (this paper)	recursive simulation of interacting country agents	economy and prices, climate and carbon, resources, society and politics, geopolitics and conflict, discrete crises and stock-flow-consistent finance	57 country agents	sectoral energy technology detail, optimal-policy search (see text)

(finding an optimal policy trajectory), GCAM and MESSAGEix on a detailed equilibrium of energy and land-use systems, and World3 on the system dynamics of global aggregates, whereas the proposed model is a recursive simulation of interacting countries with adaptive expectations. This implies a division of strengths: the assessment models and energy frameworks are *deeper* within their domains (optimal policy, sectoral energy detail), and the proposed model does not replace but complements them. Its distinction appears along three dimensions at once: (i) breadth of coupling, carrying the single loop through to society, geopolitics, and discrete crises — quantified against sectoral models in a controlled benchmark in Appendix B; (ii) uncertainty as a first-class object — ensembles and history matching are built into the workflow rather than added post hoc; and (iii) the testability of non-standard blocks (conflict-risk ranking by AUC, an identified lagged channel between the economy and society), which comparable models simply do not contain. The model thus occupies the niche of a tool for broad scenario coverage and shock-propagation analysis, enabling rather than replacing specialized models within their domains.

3. The model and its architecture

We may need to consider adding a formal uncertainty quantification specifically for growth parameters, showing how much of the ensemble spread comes from growth vs. climate parameters. Also including a scenario that does not anchor to SSP2 to show what the model

would produce endogenously. What are implications? if growth is weakly identified, are long-term policy conclusions reliable? "Elasticity of marginal utility" is mentioned in Appendix A but we don't discuss its crucial role in intergenerational welfare analysis. For a model computing the social cost of carbon, this parameter's identification may be critical.

[Addressed: growth-vs-climate decomposition of ensemble spread — Sobol analysis in Section 5 (product variance is dominated by the growth-convergence parameter, $ST \approx 0.88$, temperature variance by climate parameters, $ST \approx 0.95$); the endogenous (non-SSP2-anchored) baseline — discussed in Section 4 (emergent-scenario paragraph); the role of η — discussed with the social-cost-of-carbon results in Section 4; implications of weak growth identification — Limitations, Section 6.]

3.1. General design

The model represents the world as a system of interacting countries evolving year by year.¹ The state of each country comprises economic, resource, social, political, and military variables, as well as bilateral relations with other countries. Above the countries, the global evolution is governed by macro-variables of climate and the carbon cycle and world resource prices. One model step is one year: blocks update in a fixed order in which the output of one block serves as the input to the next, after which the year's results are passed forward to the following year. The computational core is deterministic; uncertainty is introduced through ensemble runs with random parameter draws from justified ranges (Section 4).

Below we describe the model structure decomposed into blocks grouped thematically. For each block, we indicate what it computes, on what mathematics and data it rests, and what it provides to the rest of the model.

3.2. Economy

Some subtopics which could be added 1. Technological Change and Innovation - representation of technology diffusion lags - endogenous innovation in energy technologies - learning curves for renewables. Important for climate scenarios where the cost of renewables is a key driver. 2. Submodel for pandemic. 3. Submodel for demography and migration. Add a technology learning module.

[Addressed: technology diffusion, demography-and-migration, and pandemic representation are now stated explicitly at the end of this subsection, including what is not modelled (an explicit renewables learning curve) and how the model compensates (the calibrated structural decarbonisation rate).]

Each country's output is determined by three factors: accumulated capital, employed population, and consumed energy. The production function is specified as a nested constant-elasticity-of-substitution function (a *capital-labor-energy* form), normalized so that at the calibration base point it coincides with a Cobb–Douglas production function [Arrow et al., 1961]:

$$Y = A \Theta K^\alpha L^\beta E^\gamma, \quad \alpha = 0.30, \beta = 0.60, \gamma = 0.042, \quad (1)$$

where A is total factor productivity and Θ is the technology level. The nested form, with a capital–energy elasticity of substitution $\sigma_{KE} = 0.4$ [van der Werf, 2008, Koetse et al., 2008], captures how hard it is to substitute one factor for another: as energy becomes more expensive, the economy shifts toward capital and reduces energy use. Cobb–Douglas has $\sigma = 1$. Cobb–Douglas equation but claims CES nesting. $\alpha + \beta + \gamma = 0.942$, but 5.8% is unexplained and unclear. Provide the full nested CES equation and clarify whether it's normalized to match Cobb–Douglas at the calibration point (as stated in Section 2.2).

[Addressed: the full nested form (2) and its normalization are now given explicitly; equation (1) is its base-point reduction. The exponent sum 0.942 is a deliberate mild decreasing-returns specification, and its long-run effect is bounded by a dedicated test — see the paragraph after (2).]

¹Fifty-seven actors: the 50 largest economies (Taiwan among them) and seven residual regional aggregates bundling all remaining countries of each region, so that world totals are conserved. The rationale for the cut at 50 is given with the ensemble results in Section 5.

The reviewer is right that equation (1) as written is the Cobb–Douglas limit; the production core actually computed is the nested form, of which (1) is the exact base-point reduction. Capital and energy form an inner CES composite that is then combined with labor Cobb–Douglas-style:

$$Y = A \Theta \left[\text{KE}_0 (a_K (K/K_0)^\rho + a_E (E/E_0)^\rho)^{1/\rho} \right]^{\alpha+\gamma} L^\beta, \quad \rho = \frac{\sigma_{KE} - 1}{\sigma_{KE}}, \quad (2)$$

with inner weights $a_K = \alpha/(\alpha + \gamma)$, $a_E = \gamma/(\alpha + \gamma)$, base-point quantities (K_0, E_0) captured per country at the calibration base year, and the normalization constant $\text{KE}_0 = K_0^{\alpha_K} E_0^{\alpha_E}$. At the base point the bracket equals one, so (2) coincides with the Cobb–Douglas form (1) in level *and* first derivatives for any σ_{KE} — the normalized-CES discipline that keeps the calibration point invariant while the substitution behaviour away from it is governed by $\sigma_{KE} = 0.4$ (gross complementarity between capital and energy). On the exponent sum: $\alpha + \beta + \gamma = 0.942$ encodes deliberate mild decreasing returns to scale (a proxy for fixed factors and overhead). Because each country carries a scale factor pinning its base-year output level, the exponent sum has no compounding level effect: renormalising to constant returns moves 2100 world product by only 7–9%, a bound enforced by a dedicated regression test, so no hidden 5.8% drag accumulates.

This very mechanism allows a carbon price to restructure the economy realistically. Capital accumulates by the standard rule

$$K_{t+1} = (1 - \delta) K_t + s Y_t, \quad \delta = 0.05, \quad (3)$$

where the savings rate $s \approx 0.24$ is modulated by social stability and tension — the precautionary-saving response to uncertainty [Mody et al., 2012]. Productivity A grows at a baseline rate of about one percent per year, responding additionally to research spending, technology diffusion through trade, and a conditional-convergence term in which poorer economies catch up faster — its slope was fit within the present model to the 2015–2023 cross-section of real purchasing-power-parity output to yield 0.0093 per log unit of the per-capita output gap to the frontier, so the model reproduces the observed catch-up growth of China and India. For forward projections the drift is anchored to the SSP2 socio-economic development scenario [Riahi et al., 2017]. The energy, resource, and capital markets clear through prices: energy demand and investment respond to the price of energy and the cost of capital with calibrated elasticities. Inflation (a Phillips curve with energy cost pass-through), unemployment (Okun’s law), the central-bank rate (a Taylor rule, [Taylor, 1993]), the government budget and debt, and trade flows are tracked separately. Inflation additionally carries a weak quantity-theory term: excess growth of broad money (the endogenous stock of bank deposits) above a stable-velocity reference is passed through to prices, with a pass-through coefficient ($\lambda \approx 0.027$) estimated from a cross-country panel that is small by construction — consistent with the flat post-1990 Phillips curve — and whose estimation independently confirms the model’s inflation-persistence anchor. The economic block reproduces the last decade of actual history and is therefore not speculative but empirically anchored.

Three boundary clarifications are due here. *Technological change* enters through four channels: the baseline productivity drift, an R&D-stock mechanism à la Jones fed by research spending [Jones, 1995], technology diffusion embedded in trade intensity (importers absorb productivity from more advanced partners), and the structural decarbonisation of the emission intensity of output, calibrated on the observed 2015–2023 per-country decline. An explicit endogenous learning curve for renewable-energy costs (a Wright-law module) is *not* part of the present core: the aggregate resource block does not resolve individual energy technologies, and the falling cost of clean energy is carried implicitly by the calibrated decarbonisation rate. This is an acknowledged simplification relative to energy-system models (GCAM, MESSAGEix), which resolve it explicitly — and one reason we position the model as complementing rather than replacing them. *Demography* is exogenous in the headline run (observed base-year population with scenario growth rates); a logistic demographic-transition module linking fertility decline to development is implemented and calibrated but switched off by default, following the same golden-preservation discipline as other optional mechanisms; migration responds to income gaps and instability through a gravity specification whose elasticities are anchored to the migration literature (Section 6). A dedicated *pandemic* submodel does not exist: epidemic shocks belong to the extreme-event class (discrete shocks to output, trade, and mortality) rather than to

an epidemiological compartment model, so pandemic scenarios are exercised as exogenous stress tests, not endogenous forecasts.²

3.3. Climate and the carbon cycle

Ideas of extending this subsection: 1. Specify the carbon cycle model used (with equations). 2. Include a dedicated section on how tipping points are represented (even if deactivated by default). 3. Add a comparison of the climate block's impulse response to that of MAGICC or FaIR (standard simple climate models).

[Addressed: all three added below: the pool equations (4)–(5), the tipping-point paragraph, and the FaIR/MAGICC-style ECS+TCR benchmark.]

Output is accompanied by carbon-dioxide emissions, to which a land-use source is added. The emission intensity of output declines structurally at a country-specific rate that rises with development — richer, post-industrial economies decarbonise faster (renewables and the offshoring of heavy industry), the mirror image of growth convergence and fit to the 2015–2023 per-country carbon-intensity decline.

The carbon cycle is the standard IPCC AR6 four-pool impulse-response model [Joos et al., 2013]: each year's emission pulse E_t is partitioned into four pools with shares a_i and lifetimes τ_i ,

$$C_{t+1}^{(i)} = C_t^{(i)} e^{-1/\tau_i} + a_i E_t, \quad a = (0.217, 0.224, 0.282, 0.276), \quad \tau = (\infty, 394, 36.5, 4.3) \text{ yr}, \quad (4)$$

so a fifth of every ton emitted is effectively permanent while the rest decays on centennial, decadal, and interannual timescales. Radiative forcing is logarithmic in the summed atmospheric stock, $F = 5.35 \ln(C/C_0) + F_{\text{non-CO}_2}$, where the non-CO₂ term follows the SSP2-4.5 marker pathway forward (a plateau of about 0.73 W/m² by 2100, rather than an unbounded extrapolation). Warming is computed by the standard two-reservoir surface-ocean heat-balance model [Geoffroy et al., 2013],

$$C_s \dot{T} = F - \lambda T - \kappa (T - T_d), \quad C_d \dot{T}_d = \kappa (T - T_d), \quad (5)$$

with the feedback parameter λ set by the equilibrium climate sensitivity and the heat capacities and exchange coefficient calibrated within the ranges of Geoffroy et al. [2013]. This is exactly the architecture of the reduced-complexity emulators used in the IPCC assessment (FaIR, MAGICC; Smith et al., 2018, Meinshausen et al., 2011), and the block passes the same two-number bar those emulators are constrained to: equilibrium climate sensitivity 3.0 °C (AR6 best estimate) and a transient climate response of 1.79 °C under the idealised 1 %-per-year CO₂ ramp, against the AR6 best estimate of 1.8 (likely range 1.4–2.2); both checks are pinned in the test suite.

Tipping points and carbon feedbacks. Self-amplifying carbon releases are represented by two explicit, switchable mechanisms: a proportional temperature-dependent flux into the pools (a permafrost-like feedback, GtCO₂ per degree of warming) and a stochastic abrupt-release channel whose pulse sizes follow a heavy-tailed distribution. Both are *off* in the headline run: their magnitudes are deeply uncertain, and turning them on by default would let an unvalidated mechanism dominate validated dynamics. They are exercised in tail-risk scenarios, where the carbon-feedback coefficient becomes a live parameter of the sensitivity analysis (Section 5). The block is calibrated to mainstream scientific assessment: driven by observed emissions from a pre-industrial spin-up, it tracks the observed temperature and carbon record from 1990 to 2023 with a temperature error of 0.096 °C and a best-fit equilibrium sensitivity of 3.0 °C [IPCC, 2021, Friedlingstein et al., 2023].³ This is a fast calibrated approximation, appropriate within a model of the economy and society but not a substitute for a full Earth-system model.

²Calibration sources for the economic block: the Penn World Table (factor shares and total factor productivity); the World Bank's World Development Indicators (gross product, population, energy use, military spending); the forward productivity drift is anchored to the SSP2 socio-economic development scenarios (IIASA database). A full list of variables with dataset versions and references is given in the repository file `data/external/SOURCES.md` (Section 8).

³Sources for the climate block: the equilibrium-sensitivity estimate — the IPCC Sixth Assessment Report (Working Group I); the global emissions series — the Global Carbon Project (Global Carbon Budget) and the Our World in Data compilation; the surface-temperature series for 1990–2023 — NOAA / HadCRUT5.

3.4. Warming damages and the cost of carbon

The actual damage literature has moved beyond quadratic functions. For example, papers by Marshall Burke, Stanford show non-linear, non-monotonic relationships. Reference: Burke, Marshall, et al. Quantifying climate change loss and damage consistent with a social cost of greenhouse gases. No. w31658. National Bureau of Economic Research, 2023.

[Addressed: the level-vs-growth damage discussion and the switchable Burke-type growth channel are added at the end of this subsection, citing Burke et al. [2023].]

Warming returns to the economy as damage: higher temperature reduces output. A quadratic damage function reducing the level of output is adopted [Nordhaus, 2017, Howard and Sterner, 2017],

$$\Omega(T) = 1 - 0.0078 T^2, \quad (6)$$

corresponding to losses of 3.1 %, 7.0 %, and 12.5 % of gross product at warming of +2, +3, and +4 °C. This coefficient is anchored to the productivity-corrected central estimate of the Howard–Sterner meta-analysis (about 7 % of product at +3 °C) — roughly three times higher than in DICE-2016R2 [Howard and Sterner, 2017, Burke et al., 2015] — and the damage multiplier is normalised to the observed 2023 baseline, so damage accrues only for incremental warming above the present level (no base-year double-count). A direct cross-model check isolates where our headline cost of carbon comes from: fed DICE’s *own* lower damage coefficient and discounting, the model’s marginal-pulse engine returns about \$15 per ton of carbon dioxide at a multi-century horizon — below DICE-2016R2’s \$31 per ton. That gap is not an engine artefact but a growth effect: our empirically calibrated growth core (with development convergence, Section 4) raises future per-capita consumption faster than DICE’s exogenous path, so a richer future discounts its damages more heavily. The higher headline cost of carbon therefore follows from our higher, empirically grounded damage function (about three times DICE’s), and the sub-DICE value under DICE’s own damages indicates that DICE-2016R2 underestimates damages once realistic growth is imposed. We are explicit about where this specification sits in the newer damage literature. The empirical frontier has moved beyond deterministic quadratic *level* damages toward panel-identified, non-linear responses of economic *growth* to temperature [Burke et al., 2015, 2023]: growth effects compound, so the same warming path implies much larger long-run losses, and the response is non-monotonic (cold countries can gain temporarily). The model therefore also implements a Burke-type growth-damage channel — a temperature-dependent increment to the productivity growth rate, calibrated to the panel estimates — as a switchable mechanism. It is off in the headline run for the same golden-preservation reason as other contested mechanisms: growth-effect magnitudes remain debated within the discipline, and one widely cited recent panel estimate was retracted after publication, which counsels against letting this channel drive headline results; switched on, it raises long-horizon damages and the social cost of carbon substantially, and it is explored as part of the uncertainty analysis rather than hidden. The quadratic level function (6), anchored at the meta-analytic central estimate, remains the auditable, conservative headline choice. From the damage function, the social cost of carbon — the economic damage from emitting one additional ton of carbon dioxide, a standard reference point for climate policy — is computed by the marginal-pulse method, with a FaIR-style pulse injected into the carbon pools of (4). This quantity depends substantially on the discounting scheme and on the structure of future growth; results are reported in Section 4.⁴

3.5. Resources

For each country, stocks, production, and consumption of energy, food, and metals are tracked; world prices rise when demand exceeds supply, and this price is passed into inflation and on into the economy. The resource market clears by price with a calibrated demand elasticity; a slow-adjustment rule is provided as a fallback. Resources are represented in aggregate rather than as detailed sectoral energy systems.⁵

⁴Calibration of the damage function draws on the Howard–Sterner meta-analysis and the panel estimates of Burke and coauthors; the social-cost-of-carbon reference points are the estimates of the U.S. Environmental Protection Agency (EPA, 2023) and the RFF-SP project.

⁵The resource block is specified in aggregate: energy indicators draw on the World Development Indicators; price elasticities are set from generalized market estimates rather than a dedicated sectoral dataset.

3.6. Society, politics, geopolitics, and culture

Specifying how political regime type affects economic outcomes and policy choice may help. Are institutional quality endogenous? Can it decline due to economic stress? How does regime type affect policy responses?

[Addressed: the operational definition of social tension, the endogenous-legitimacy mechanism, and the explicit regime-type effects are now spelled out in the three paragraphs added at the end of this subsection, together with an honest statement on heterogeneous political economies (state enterprises, capital controls).]

For each country the model tracks trust in government, social tension, income inequality, institutional legitimacy, and protest pressure. These variables respond to the economy — rising inflation and unemployment raise tension and erode trust, and high inequality amplifies this effect — and, in turn, affect stability and crisis risk. Each pair of countries is linked by a relation: trade volume, the presence of an alliance, the level of conflict or tension, a state of war, and sanctions. Each country's military capability is anchored to observed measures of national power rather than to arbitrarily assigned values [Singer et al., 1972]: a four-component CINC-style capability share (population, energy, gross product, and SIPRI military expenditure, the latter populated at world build from a committed SIPRI-2023 grounding file). The military-expenditure component matters empirically: the three-component proxy without it reproduces the cross-country *levels* of military spending well (correlation ≈ 0.76) but is anti-correlated with the post-2022 militarization *dynamics* (correlation of share changes -0.115), so no combination of population, energy, and output carries the militarization signal. A small set of cultural traits (chiefly individualism, uncertainty avoidance, long-term orientation, and hierarchy), together with the character of the political regime, influence economic and social behavior [Hofstede et al., 2010]. Unlike the economy and climate, no single authoritative calibration dataset exists for these links; where possible their priors are tied to the literature and to reproducible numbers (Section 5), and the remainder is expert-set and assessed against a more modest criterion (Section 4).⁶ Traits that did not in fact affect outcomes have been removed from the model, so that it does not mimic cultural detail it does not use.

Operational definition of social tension. Tension is a latent index on $[0, 1]$, but its construction is fully specified and cross-country comparable. At the base year it is a fixed-weight composite of observed quantities: $0.35(1 - \text{trust in government}) + 0.25 \text{ Gini (normalized)} + 0.15 \text{ water stress} + 0.10 \text{ climate risk} + 0.15(1 - \text{regime stability})$, where trust and regime stability are themselves fixed-weight composites of Worldwide Governance Indicators rescaled from their published ± 2.5 scale to $[0, 1]$ — the same units for every country, which is what makes levels comparable. Thereafter it evolves by increments: unemployment adds 0.010 per point, inflation 0.005 per point, inequality contributes in proportion to the Gini index damped by cultural individualism, a mean-reversion term pulls toward the trust anchor, long-term orientation damps the total, and a small spillover term (0.05) couples a country to the mean tension of its geographic neighbours. All coefficients carry priors varied in the ensembles (Appendix A); the index is validated by channel (Section 5) and by its contribution to conflict ranking (Section 4), not by pretending a direct observable exists.

Institutions and regime type. Institutional quality is endogenous and can decline under economic stress: legitimacy is recomputed every year as $0.6 \text{ trust} + 0.4(1 - \text{tension})$, and, together with protest pressure and debt stress, it determines a country's *policy space* — a multiplier that scales down the magnitude of all feasible policy actions as legitimacy erodes; crises knock regime stability down directly. Regime type (democracy, autocracy, hybrid) is a fixed categorical attribute that modulates responses: in democracies the social-tension response to burden-shifting measures such as fuel taxation is amplified ($\times 1.2$) while autocracies damp it ($\times 0.8$), and under external threat autocracies exhibit a rally-round-the-flag trust gain while other regimes lose trust. What the model does *not* do is transition regime type endogenously: a regime-collapse crisis resets state variables but not the categorical type.

Heterogeneous political economies. Countries with state-dominated economies and capital controls (China, Russia, Iran, and others) are represented by the same functional forms with country-specific

⁶The socio-political and geopolitical variables are anchored to recognized indices: income inequality — the SWIID database and the World Bank Gini index; legitimacy, governance quality, and political stability — the Worldwide Governance Indicators; armed conflicts — the UCDP/PRIO Armed Conflict Dataset (v. 24.1); military capability — a CINC-style composite (Correlates of War methodology) whose military-expenditure component is populated from the SIPRI military-expenditure database (2023 values, committed as `data/external/sipri_milex_2023.csv`); sanctions — the Global Sanctions Database (v. 3); cultural traits — the Hofstede indices. The mapping of each variable to its source is described in `data/external/SOURCES.md`.

states — governance indices, inequality, cultural traits, military expenditure, regime type — rather than by structurally distinct equations. The regime-type modifiers above carry part of the difference, and the empirical anchoring (the model reproduces China’s and India’s observed catch-up growth) shows the shared forms are not grossly wrong on the economic side. But state-owned-enterprise credit allocation, administered prices, and capital controls are not structurally modelled; this is a stated limitation (Section 6), mitigated in scenario work by the language-model policy layer, which can impose country-specific doctrines on top of the shared mechanics.

3.7. Risk, crises, and finance

A few questions on the finance part: 1. Can the central bank set interest rates independently? 2. How are exchange rates determined? 3. How does sovereign debt restructuring work in the model? 4. How international capital flows are included? The full financial sector submodel with equations may need to be added.

[Addressed: all four questions are answered in the new “monetary and external side” paragraph at the end of this subsection: per-country Taylor rules; no nominal exchange rates (real PPP numeraire) with an explicit reserve-based currency-crisis mechanism instead; the full restructuring mechanics (40 % haircut, output and persistence multipliers, exit rule); and reserve-settled trade in place of a full capital account.]

On top of the smooth annual dynamics, the model recognizes and applies discrete crises — debt, currency, political-regime collapse, and extreme climate events — each triggered when the corresponding threshold is crossed. Crisis severity may be drawn from a heavy-tailed distribution (most events are moderate, rare ones catastrophic), matching the observed distribution of real crises and wars [Taleb, 2007, Weitzman, 2009]; this mode is optional and is explored as uncertainty. The financial side is kept with strict accounting: every change in debt is balanced by a counter-flow. Government finances are closed, and the private sector is described by a full bank balance sheet in which every loan creates a matching deposit, so that the money stock is identically equal to deposits and loans; excessive borrowing raises the cost of credit and restrains the economy — a financial accelerator [Bernanke et al., 1999, Godley and Lavoie, 2007].⁷

The monetary and external side, concretely. Each country runs an independent central bank: the policy rate follows a Taylor rule with weights 0.5 on the inflation gap and 0.5 on the output gap (via the unemployment gap to its natural rate), capped at ± 6 percentage points around the neutral rate [Taylor, 1993]. Nominal exchange rates are *not* modelled: all quantities are carried in a common real purchasing-power numeraire, so “currency crisis” is triggered not by a depreciation but by its fundamentals — simultaneously external debt above 50 % of output, a current-account deficit beyond 4 %, and foreign-exchange reserves below three months of imports — the classic first-generation ingredients. Sovereign distress has explicit mechanics: a debt crisis fires when debt exceeds 120 % of output *and* the effective rate — the policy rate plus a debt-dependent sovereign spread whose linear and quadratic coefficients are anchored to the sovereign-risk literature [Hilscher and Nosbusch, 2010, Arora and Cerisola, 2001] — exceeds 12 %; the crisis year applies a 40 % debt write-down (restructuring) and a 7 % output loss, unresolved crises decay output further each year, and exit requires debt below 70 % of output at a rate below 8 %, within at most six years. International capital flows are represented at the level the model needs and no further: bilateral trade settles in foreign-exchange reserves (with bounded trade credit), net exports are tracked per country and sum to zero across the world by an enforced invariant, and reserve depletion feeds the currency-crisis trigger; a full portfolio capital account (cross-border bond and equity holdings) is not modelled and is listed among the limitations.

3.8. Block interactions

Connectivity is the governing property of the model. The economy generates emissions; emissions determine warming; warming, through the damage function (6), reduces output; the carbon price raises the cost of energy and, through substitution in (1), lowers emissions. Economic hardship raises social tension and lowers trust, which changes the savings rate in (3) and raises crisis risk; a crisis hits output and capital

⁷There is no dedicated calibration dataset for the risk-and-finance block: the heavy-tailed crisis-severity distributions and the stock-flow-consistent accounting rest on the methodological literature cited in the text rather than on a single external source.

Table 2

Empirical basis of the model's blocks.

Block	Data source	Purpose
Economy	Penn World Table, World Development Indicators	factor shares, growth, debt
Climate and carbon	IPCC AR6 assessment, Global Carbon Budget	sensitivity, T and CO ₂ paths
Cost of carbon	EPA-2023, RFF-SP	reference for cross-checking
Forward growth	SSP2 scenarios	forward productivity drift
Conflict	UCDP Georeferenced Event Dataset	conflict-risk validation
Inequality	SWIID	income inequality
Institutions	Worldwide Governance Indicators	legitimacy, governance quality
Military power	CINC-style composite incl. SIPRI military expenditure	country capability
Culture	Hofstede indices	cultural traits

and is transmitted to trading partners. Resource scarcity raises prices and inflation; conflicts and sanctions disrupt trade. Trade is not a symmetric abstraction: bilateral flows are settled in reserves, each country's net exports are tracked explicitly (the world sums to zero by an enforced accounting invariant), and persistent deficits drain the reserves and raise the external-debt burden that feed the currency-crisis trigger — so trade imbalances are a transmission channel, not an untracked residual. A shock in one subsystem thus propagates through the others — and this coupling is exactly what gives the model its substantive value for risk analysis. This inter-subsystem network is geographically grounded: bilateral trade follows a gravity law (intensity falling with distance, elasticity ≈ 0.9 ; Head and Mayer, 2014), and conflict, social unrest, and climate risk spill over to bordering neighbours, so a shock propagates along realistic spatial pathways rather than uniformly. Activating this geographic coupling leaves the 2015–2023 retrospective fit essentially unchanged (world-product, CO₂, and temperature RMSE move by under a quarter of a percent), so the gain in propagation realism costs nothing in validated accuracy.

4. Calibration and validation

We may need to add how thresholds are set for crises (debt, regime collapse). From literature, calibration, or expert judgment? How the "conflict-proneness score" is constructed from the model's internal variables?

[Addressed: crisis-threshold provenance is now stated at the end of the calibration-principles subsection (literature-anchored expert priors, tagged as such in the parameter registry, with the sovereign-spread block empirically anchored); the conflict-proneness construction is given explicitly in the conflict paragraph of Section 4.]

4.1. Data sources

Each block is anchored to external data (Table 2). The economic and climate blocks are calibrated and validated most rigorously; the socio-political and geopolitical variables are anchored to recognized indices but assessed against a softer criterion.

4.2. Calibration principles and integrity control

Calibration follows several rules. First, every optional mechanism is implemented as a switch, off by default unless anchored by historical evidence; thus the model's headline ("base") behavior remains tied to data, while speculative mechanisms are explored separately as uncertainty. Second, parameter estimation uses not point fitting but *history matching*: instead of seeking a single "best" parameter set, regions of parameter space incompatible with observations are discarded, leaving a posterior region of "not-yet-ruled-out" values — a standard approach for computationally expensive simulation models [Williamson et al., 2013]. Concretely, the constraining observations are the 2015–2023 retrospective series of world product, world emissions, and temperature. For each candidate parameter vector θ an implausibility measure is

computed per observable, $I_o(\theta) = \text{RMSE}_o(\theta)/\text{tol}_o$, where the tolerances (6 trillion USD, 2 Gt, 0.15°C) bundle observational uncertainty with an explicit model-discrepancy allowance; a vector is retained in the not-yet-ruled-out (NROY) region if $\max_o I_o \leq 3$ — the conventional three-sigma cut. The retained region then tightens the priors used by the ensembles. Because the core is deterministic, a single simulation per parameter vector suffices; when internal temperature variability is switched on, the temperature criterion is evaluated over an eight-seed mini-ensemble so that the cut applies to the forced response rather than to noise. Exterior cross-checks (the published range of the social cost of carbon, the conflict-registry ranking skill) act as additional one-sided constraints. Third, the climate coefficients derived from the initial state are tightly bound to a checksum of the input data and cannot be changed by hand; the integrity of this link is checked automatically. Finally, regression control is in force: the significant retrospective-validation quantities are pinned, and any change to the core must either preserve them within tolerance or be accompanied by an explicit recalibration.

Crisis thresholds deserve a provenance note of their own, since no clean estimation dataset exists for them. The trigger values — debt crisis at debt above 120 % of output with an effective rate above 12 %; currency crisis at external debt above 50 %, a current-account deficit beyond 4 %, and reserve cover under three months; regime collapse at trust below 0.20 with tension above 0.80 — are expert priors informed by the standard reference points (the Maastricht debt criterion, the Reinhart–Rogoff high-debt literature, first-generation currency-crisis models) and are tagged **PRIOR** in the parameter registry, distinguishing them from the empirically fitted **DATA** entries such as the sovereign-spread coefficients; the severity and persistence multipliers around them are calibrated to the plateau behaviour of the retrospective window. The registry makes this distinction auditable parameter by parameter.

4.3. Resulting metrics

Ideas of strengthening this subsection: 1. Report validation against the World3 model or other integrated models, not just persistence/linear forecasts. 2. For conflict prediction, report calibration metrics (reliability diagrams), not just discrimination (AUC). 3. Show the actual predicted vs. observed conflict counts, not just ranking.

[Addressed: (1) a direct trajectory comparison against World3 is not meaningful — it models global aggregates without a price economy, countries, or a modern climate module — so the cross-model checks used instead are the DICE social-cost-of-carbon engine swap in this section and the sectoral-model benchmark of Appendix B; (2) a calibration-style read (monotone conflict incidence by score quintile) now accompanies the AUC; (3) observed counts by ranking bucket (top/bottom twenty) are now reported in the conflict paragraph.]

A validation summary across all blocks is given in Table 3, with the key results discussed below. Retrospective validation over 2015–2023 yields root-mean-square errors for gross product, world emissions, and temperature. With the corrected capital initialisation both the Cobb–Douglas and the closed nested-substitution core reach a product error of about 0.60 trillion USD (normalising the damage function to the 2023 baseline removed a base-year double-count and improved this fit); the gain from the nested substitution function with market clearing shows in emissions (Fig. 2a), where it lowers the error from 1.74 to 0.94 gigatons. The temperature error of 0.145°C is reported with the internal-variability spread re-derived from the observed 1990–2023 forced residuals ($\sigma = 0.088$, near-white), which also cures an ensemble under-dispersion present in earlier versions.

The Brier Skill Score (BSS) is reported as +0.143, which is barely positive. Many forecasters would consider BSS < 0.1 as "no skill."

[Addressed: agreed that the BSS is modest (and it is +0.123 under the v18 re-anchor); the conflict paragraph now adds the quintile calibration read (monotone 55 % → 8 % incidence), the top/bottom-20 counts, and the framing argument: zero parameters are fitted to the conflict registry, so any positive out-of-sample skill is informative — the claim defended is relative risk ranking, not event prediction.]

Comparison with benchmark forecasts reveals the structure of this accuracy. The model *beats* the persistence forecast for temperature (a skill score of about +0.14) and marginally for product, but *loses* to a naive linear extrapolation for product and emissions over this short, smooth interval. This is to be expected and is consistent with assessments of professional forecasts [Celasun et al., 2021]: a structural

Table 3

Validation summary by model block: data source, validation criterion, and the achieved result. The economic and climate blocks are checked by strict retrospective validation; for the others the criterion is softer — cross-checking against published estimates, improvement over a naive benchmark, or channel identification. The comparison of the current core with the former one (Cobb–Douglas) is shown in Fig. 2a.

Block	Source	Criterion	Result
Economy	PWT, WDI	retrospective RMSE, 2015–2023	GDP 0.60 trillion USD (Cobb–Douglas 0.60)
Climate and carbon Cost of carbon	IPCC AR6, GCB EPA-2023, RFF-SP	retrospective RMSE, 2015–2023 published cross-check; Ramsey discounting	CO ₂ 0.94 Gt; T 0.145 °C ≈\$90/t CO ₂ (range 90–280); under DICE's lower damages SCC is ≈\$15, i.e. DICE-2016R2 underestimates damages relative to our empirically calibrated function
Conflict	UCDP/PRIO	ranking vs. registry	AUC 0.739 [0.60; 0.86]; BSS +0.123; $p \approx 0.001$
Social block	SWIID, WGI	channel identification (Sec. 5)	significant lagged channel; regression $p < 10^{-12}$

model raises skill where dynamics matter (climate) and loses to trivial extrapolation where a short-run smooth aggregate is hard to beat.

What does the model produce *without* the SSP2 anchor? The anchor enters only the forward productivity drift; emissions, decarbonisation, and the climate response are endogenous. Letting the model run on its own mechanics, the emergent no-intervention baseline lands nearest the *strong-mitigation* end of the assessed scenario space (about 2 °C of warming by 2100, closest to the SSP1-2.6 envelope): the structural decarbonisation rate fitted to 2015–2023 is optimistic relative to an SSP2-style reference world when extrapolated for decades. We report this as a finding with a practical implication: absolute long-horizon levels inherit the weak identification of long-run growth and decarbonisation (Section 6), which is precisely why the recommended use mode is scenario *differences* under a common core (Section 7), where this anchor uncertainty largely cancels.

A different criterion is applied to the conflict block. The countries' conflict-proneness score is a fixed-weight composite of observed inputs — 0.35 (1 – regime stability) + 0.25 social tension + 0.20 climate risk + 0.20 military burden (the military-expenditure share of output, robustly normalized) — with no coefficient estimated from conflict data. It is compared with the historical registry of armed conflicts over 1990–2023 across 57 countries [Sundberg and Melander, 2013]. The framing of the test is essential: what is evaluated is not an output fitted to these same data but an *input* risk indicator computed from observed country characteristics and not tuned to the conflict registry. Hence there are no parameters trained on the target variable and no associated information leakage, and a rolling train/test split is unnecessary; the full set of 57 countries is evaluated as a whole. The model confidently ranks conflict-affected countries above calm ones: the area under the characteristic curve is 0.739 (bootstrap 95 % interval [0.60; 0.86], 20 000 pair resamples with replacement [Efron and Tibshirani, 1993]), and a permutation significance test (random label shuffling, 50 000 repetitions) gives $p \approx 0.001$, so the ranking is decisively better than chance. The Brier skill score relative to the base-rate forecast is +0.123 (a positive value indicates predictive value). Discrimination is complemented by a calibration-style read of the same ranking: splitting the 57 actors into score quintiles, the observed 1990–2023 conflict incidence falls monotonically from 55 % in the top quintile to 8 % in the bottom; of the twenty lowest-ranked actors only three saw conflict, against eleven of the top twenty. On the magnitude of the skill scores: a Brier skill of +0.12 against the base rate is indeed modest, and we make no stronger claim — but the appropriate comparison class is conflict forecasting, where even purpose-built models trained on the target registry with rich event data struggle to beat simple baselines out-of-sample, and where climate–conflict links remain contested [Hsiang et al., 2013]. The score here is an *input* indicator with zero parameters fitted to conflict outcomes; that it nonetheless ranks (AUC 0.739, $p \approx 0.001$) and calibrates (monotone quintiles) against the registry is evidence the model's internal state carries real conflict-relevant

signal. This is a reasonable bar for a block of this kind: the model is suitable for assessing relative risk but not for predicting specific wars on specific dates.

The social cost of carbon is sensitive to the discounting scheme and to the integration horizon. Under a modern Ramsey scheme with a near-zero pure rate of time preference it is roughly \$90 per ton of CO₂ at a two-century horizon — within the broad modern range, below the central EPA-2023 and RFF-SP estimates (about \$190 per ton of CO₂) [Rennert et al., 2022, U.S. Environmental Protection Agency, 2023]; under Nordhaus-style discounting it is about \$22, \$43, and \$46 per ton at the 30-, 100-, and 200-year horizons (the 30-year value truncates the long-run damage tail — the honest horizon comparison). Importantly, this quantity is *genuinely* sensitive to the structure of future growth: across different calibrations of the economic core it varies over a range of roughly \$90–280 per ton of CO₂, and the development-convergence recalibration of the growth core (faster catch-up growth, which makes the future richer and discounts its damages more heavily) moved the central value down by about a quarter. Fed DICE-2016R2’s own, lower damage coefficient our pulse engine returns about \$15 under this faster growth — below DICE’s \$31 per ton of CO₂ — so DICE underestimates damages relative to the empirically calibrated function we adopt. We report these values together with their sensitivity rather than tuning to a target; that sensitivity is itself a standalone result. Two welfare parameters govern the discounting side: the pure rate of time preference ρ and the elasticity of marginal utility η , which in the Ramsey rule $r = \rho + \eta g$ converts the growth rate g of consumption into the discount rate and thereby encodes intergenerational inequality aversion. Neither is identified by historical data inside the model — they are ethical primitives, not calibration targets — so we treat them exactly as the climate-economics literature does: η carries a prior of 1.0–2.0 around a central 1.45 (Appendix A), and the reported \$90–280 span of the social cost of carbon is dominated by the joint (ρ, η) choice interacting with the growth path — the faster the convergence-driven growth, the harder a high η discounts future damages. Reporting the cost of carbon conditional on the discounting stance, rather than as a single number, is therefore a design decision, not an evasion.

5. Uncertainty, sensitivity, and parameter identification

A key question is whether the model’s many parameters are identified by data rather than set by hand. Each key parameter is equipped with a prior distribution — a central value, a plausibility range, and a literature source (Appendix A); estimation proceeds not by point fitting but by history matching (Section 4), discarding regions of parameter space incompatible with observations. Thus the capital–energy elasticity of substitution $\sigma_{KE} = 0.4$ rests on empirical estimates in the 0.3–0.5 range, the depreciation rate $\delta = 0.05$ on Penn World Table data (3–8%), and the damage coefficient 0.0078 sits within the empirical range 0.0015–0.025, at the productivity-corrected central estimate of the Howard–Stern meta-analysis and well above the DICE-2016R2 value. The irreducible uncertainty of these quantities is not hidden but carried into the computation through ensembles.

Sensitivity analysis. To establish which parameters actually determine outcomes, a global screening by the Morris elementary-effects method [Morris, 1991, Campolongo et al., 2007] was carried out over 33 key parameters for four output quantities (Fig. 3; 8 trajectories, 10-year horizon). The picture is consistent with the model’s structure and with physics: temperature is determined almost exclusively by climate parameters — equilibrium sensitivity, ocean heat exchange, and surface heat capacity; world product by capital depreciation, the energy intensity of output, and the damage coefficient; world emissions by the emissions scale, the land-use source, and the capital stock that scales activity. These conclusions are robust to the screening budget: increasing the number of trajectories from 8 to 32, the Spearman rank correlation of the full μ^* vector with the reference run is $\rho \approx 0.99$ for the three physical quantities (in the range 0.986–0.995), and the dominant parameters reappear across all tested random-number seeds: a complete top-five match for temperature, and for emissions and product a stable dominant triple (emissions scale, land use, and capital depreciation for emissions; capital depreciation, energy intensity, and the damage coefficient for product), while the lower-ranked parameters sit close together and may exchange places — consistent with the Sobol finding that the structural decarbonization rate, once re-centred on the observed 2015–2023 decline, contributes little residual variance. Social tension, by contrast, is almost insensitive to these physico-economic parameters: its elementary effects lie at the screening noise floor (the absolute μ^* is about three orders of magnitude smaller than for the physical quantities), and their ranking — unlike

the stable ranking of the physical quantities — is itself unstable to the choice of seed ($\rho \approx 0.93$ at $8 \rightarrow 32$ trajectories, versus ≈ 0.99 for the physical quantities). Importantly, however, this does *not* mean that society is decoupled from the economy: this screening varies only the physico-economic priors, whereas tension is governed by its own (social) parameters and responds to economic conditions with a multi-year lag. We isolate this channel from economic conditions to social tension separately below (Fig. 5). The screening thus confirms that the significant quantities are governed by substantively appropriate and empirically anchored parameters, and this conclusion is not an artifact of the limited computational budget. The Morris screening is complemented by a variance-based Sobol–Saltelli analysis over the twenty most influential priors: output variance is concentrated and attributable (equilibrium climate sensitivity dominates temperature, catch-up convergence dominates product, sovereign-spread coefficients dominate inequality), and ten of the lower-ranked priors can be frozen without materially changing the output distributions — evidence that the effective dimensionality of the calibration is far smaller than the parameter count.

Ensemble uncertainty quantification. Because the core is deterministic, uncertainty is introduced by running a Monte Carlo ensemble (500 members) with random parameter draws from their prior ranges. The result is not a single trajectory but a distribution of outcomes (Fig. 4). Throughout, “world product” denotes the *aggregate gross product of the 57 modeled actors* — the 50 largest economies (Taiwan among them) plus seven residual regional aggregates that bundle all remaining countries of each region — so world totals are conserved by construction; in the 2023 base year (the validated canon) it is about 107 trillion USD. The cut at 50 is a coverage threshold, not a convenience: the direct countries carry about 93.5% of world product, the smallest of them contributes only $\sim 0.25\%$, so each economy below the cut is individually negligible for global dynamics and is carried by its regional aggregate instead — while 57 actors keep a full bilateral relation matrix (57×56 directed pairs) cheap enough for ensemble work. At the ten-year horizon its median is about 136 trillion USD, with an interquartile range of roughly 134–139 and a 5–95th-percentile interval of roughly 131–142; the 5–95th-percentile interval of the temperature anomaly is roughly $1.1\text{--}2.0^\circ\text{C}$ and reflects chiefly the spread in equilibrium sensitivity. The reported percentiles have converged in ensemble size: as the number of members grows from 80 to 500, the median and upper percentiles of product and temperature are stable (the product median moves by 0.5 trillion USD, about 0.4%, and the upper percentiles by less; the temperature median by 0.05°C); the only quantity noticeably dependent on sample size is the lower temperature tail, which tightens from ≈ 1.05 to $\approx 1.1^\circ\text{C}$. It is precisely such intervals, rather than point values, that constitute the model’s substantive result. Two design caveats are stated rather than hidden. Priors are drawn *independently*: cross-parameter correlations (for example, between climate sensitivity and ocean heat uptake, which are anticorrelated in Earth-system calibrations) are not imposed, so the ensemble slightly overstates spread where such correlations would constrain joint draws — a conservative direction for a risk tool. And convergence is diagnosed empirically (the $80 \rightarrow 500$ member comparison above) rather than assumed; percentile stability, not member count, is the reported criterion.

The channel from the economy to social tension. The low sensitivity of tension to the physico-economic *priors* (above) does not mean that society is decoupled from the economy: tension is an accumulating variable (its increment is determined by unemployment, inflation, and inequality), so its link to the economy manifests not instantaneously but with a lag, and is invisible to a fixed-horizon screening. We isolate this channel through three complementary checks (Fig. 5; the reproducible computation is described in Section 8). First, *localization*: in an ensemble that additionally varies the priors of the social block itself, final tension is significantly governed by them — by the sensitivity to inequality (Spearman rank correlation $\rho \approx +0.59$, $p < 10^{-11}$) and the strength of the trust anchor ($\rho \approx -0.52$, $p < 10^{-9}$) — whereas the physico-economic priors remain weak; that is, the screening above simply did not vary the block’s direct drivers. Second, *accumulation*: the prior-induced spread of tension grows with the horizon (the across-member standard deviation increases by roughly two thirds from 5 to 25 years), which is the signature of a lag. Third, *impulse response*: an exogenous stagflation shock (a three-year rise in unemployment and inflation) raises mean tension persistently, with the response still building a decade after the shock (positive in 78% of the prior-ensemble runs at its peak); the spread grows with horizon because in some runs the shock triggers discrete crises. A distributed-lag regression of the change in tension on economic stress (within-member fixed effects, clustered errors) confirms significant contemporaneous transmission (lag 0: $p < 10^{-8}$), with a positive cumulative effect over five lags. The channel from the economy to tension is thus statistically significant but

mediated and lagged; it is precisely this lag, not an absence of any link, that explains the null result of the short-horizon screening.

Identification of the military-capability grounding. The same discipline — test the input against data it was not tuned to — is applied to the military block. On a 48-country panel of military expenditure over 1995–2024 (country fixed effects, errors clustered by country), peacetime spending growth tracks output growth nearly one for one (coefficient 0.79, $p < 0.001$), so in the normal regime military expenditure is an economic phenomenon; the post-2022 excess over the economic prediction, however, is structured by geography rather than by income — about +13 percentage points per year for the NATO frontier states, decaying to approximately zero for the rest of the world — while a classic Richardson action–reaction term (chasing the rival’s spending growth rate) is decisively rejected ($p = 0.74$). This motivates the grounding adopted in Section 2.6: military expenditure enters the capability index as an observed state variable rather than through an autonomous arms-race equation, and the threat-dependent regime shift is carried by the dyadic relation variables that already represent it.

Computational cost. All results in this paper are computed on a single consumer laptop (Apple M4 Pro, 12 cores, 24 GB): one 57-actor world-year costs 30–50 ms, so a deterministic ten-year world run takes well under a second; the 500-member ten-year Monte Carlo ensemble completes in about 47 s across cores; one Morris screening pass (33 parameters, 8 trajectories) takes 2–3 minutes per output quantity; the Sobol–Saltelli analysis (2,816 deterministic 2015–2050 runs) about half an hour; and the full 527-test regression suite, including the golden backtests, about two minutes. No cluster hardware is required anywhere in the workflow, which is what makes ensemble-first uncertainty quantification practical as a default mode rather than a special exercise.

6. Limitations

Add a "Case Study" section with a policy-relevant question answered using the full ensemble and weak-signal tools. Clarify the socio-political block with transparent protocols. Revisit the ensemble design with proper treatment of parameter correlations and convergence diagnostics. Present the validation metrics with full disclosure of where the model fails. Expand LLM integration section to be scientifically reproducible.

[Addressed: worked policy case — Section 7 (carbon-tax case study) with the full dose–response harness in Appendix B; socio-political protocols — operational definition of tension in Section 2.6 and the anchored-vs-expert ledger in Section 6; parameter correlations and convergence — the ensemble-design paragraph in Section 5; failure disclosure — Section 4 (the model loses to linear extrapolation on short smooth aggregates) and the saturation caveat in Appendix B; LLM reproducibility — the expanded protocol in Section 8.]

An honest list of what the model *cannot* yet do is no less important than a list of its capabilities.

Expectations in the headline run are adaptive. Households and firms look to the recent past rather than to a rationally anticipated future, so the headline run does not reproduce anticipatory effects (for example, a response to a pre-announced policy before it takes effect). A model-consistent (near-rational) expectations mechanism has been implemented and tested, but an ensemble check found its marginal effect statistically unstable, so it is left off by default.

Long-run growth is weakly identified. The long-run rate of productivity growth is generated by an endogenous mechanism (R&D-stock accumulation à la Jones, technology diffusion, frontier convergence) but is only weakly pinned down by the available short history and, for forward projections, is anchored to an external scenario rather than derived from it.

The climate is described by a fast approximation. The climate block reproduces the standard indicators but is not a full Earth-system model; the strongest carbon feedbacks are deeply uncertain and are relegated to tail scenarios.

Damage estimates are uncertain across the discipline. Estimates of climate damage are uncertain across the entire field; the model carries this uncertainty as an interval rather than as a single number.

The socio-political links are partly anchored, partly expert. There is no single calibration dataset for the transition from conditions to discontent, but a substantial part of the social-block priors is now tied to the literature and to reproducible numbers (Section 5): the war-size power-law tail exponent, the migration-gravity elasticities, the regime-collapse GDP-loss magnitude, and conflict-onset forecast skill —

and a geographic coupling checked against the spatial dependence of the real record. What remains expert-set are the coupling thresholds and, in particular, the weight of cross-border tension diffusion, for which there is no clean adjacency anchor: in the most careful cross-national study, pure-adjacency protest spillover is statistically insignificant. The channel from the economy to tension is nonetheless significant, directionally correct, and lagged (Fig. 5); conflict is intrinsically hard to predict, and the geopolitical block is suitable only for relative risk (AUC ≈ 0.74).

Crisis triggers are threshold-based. Discrete crises are launched by threshold rules and capture the correct qualitative behavior without being derived from a single fitted crisis dataset; their severity distribution, however, is anchored to the power-law of war sizes (exponent $\alpha \approx 1.5$; Richardson; Clauset).

The geographic coupling is realistic but its payoff is concentrated in trade. Shock propagation is grounded on a real spatial graph, and the model reproduces the spatial dependence of the real record (a neighbour-conflict premium; an emergent Moran's I). The substantive payoff, however, is concentrated in the trade channel (anchored gravity), while the cross-border diffusion weights for conflict, tension, and climate risk are modest and, for tension, lack a clean literature anchor.

The overall stance is unchanged: the realistic goal is to be calibrated against one's own uncertainty rather than to serve as an instrument of infallible foresight. A realistic accuracy bar for multi-year forecasts is improvement over a naive benchmark.

7. Typical use cases

The model is intended for three interrelated modes of use, in which its connectivity and proper uncertainty quantification pay off most.

Scenario analysis and strategic planning. The principal mode is comparison of a baseline trajectory with a perturbation or policy scenario: a shock or measure is introduced (for example, a carbon price, resource scarcity, or an escalation of relations between countries), and its propagation through the economy, climate, society, and geopolitics is traced. This makes it possible to compare alternative strategies and assess the attendant risks. What matter here are *relative* quantities — differences between scenario and baseline — rather than absolute point values at a distant horizon.

A worked policy case. “What does a \$50/tCO₂ carbon tax buy, and what does it cost politically?” is answered end-to-end by the machinery of this paper: against the baseline, the tax lowers world emissions by about 5.5 % over ten years (the first-order effect any integrated assessment model reports) while raising social tension and lowering trust in government through the energy-price-inflation channel — a cost invisible to single-domain models, monotone in the tax rate, and directly relevant to the political sustainability of carbon pricing (the French 2018 and Australian 2014 reversals). The full dose-response computation, including the oil-shock and crop-failure counterparts, constitutes Appendix B; run as ensembles, each arm carries its percentile band, and comparing the weak-signal profile of a shocked arm with the baseline's (Section 7) is the intended early-warning reading of such a case.

Ensemble computations. Because the computational core is deterministic, uncertainty is introduced by running large ensembles with random parameter draws from justified ranges. This yields not a single trajectory but a distribution of outcomes, against which sensitivity to each parameter is measured and probabilistic estimates are constructed. Scenarios are best compared as ensembles rather than as single runs.

Weak-signal detection. The built-in crisis logic is threshold-based: a crisis fires when a boundary is crossed. Weak-signal detection complements it: the identification of a subtle, multivariate accumulation of tension *preceding* a structural shift, before any threshold is crossed. Three complementary tools are used. The first is the Mahalanobis distance: at each point in time it is assessed how far the *joint* state (product, temperature, debt, tension, and so on, together) lies from its recent baseline distribution, accounting for correlations between dimensions; exceeding a χ^2 threshold indicates the system's transition into an unusual region of state space [Mahalanobis, 1936, Wilson and Hilferty, 1931]. The second is structural-break detection: a single series is searched for a regime change by change-point detection with an information-criterion penalty, yielding a probability of a shift and its most likely location [Schwarz, 1978]. The third is critical-slowness indicators (rising autocorrelation and variance) that grow as a bifurcation point is approached [Scheffer et al., 2009]. So that these tools detect deviations from the *dynamics* rather than secondarily detecting a smooth

trend, the series are first reduced to stationary increments. On synthetic data and on the model’s own trajectories the tools behave as expected: an injected 15 % drop in product is detected exactly at the right step, whereas a calm run stays near the theoretical false-alarm rate. The working device here, as in scenario analysis, is *relative*: comparing a scenario’s weak-signal profile with the baseline’s (earlier and more frequent deviations, a higher shift probability) indicates that the scenario is moving the system toward a structural transition.

8. Software implementation and the scenario environment

The model is implemented as an open codebase and is available in the repository <https://github.com/Theclimateguy/GIM>; the results reported here correspond to release v18.1.3. The computational core is deterministic and reproducible: each run is written to a separate dated folder with a manifest of parameters and input-data versions, and a suite of unit tests and a validation protocol pin the key retrospective-validation quantities (Section 4). Calibration datasets are attached as immutable, checksum-bound inputs, which precludes silent divergence of results between runs. The baseline world run, scenario questions, multi-agent games, and calibration are launched through a single command-line interface; the engine is also packaged as a pip-installable library with the runtime data bundled inside the package.

Large language models as countries. Some questions to be considered here: What LLM is used? How are state variables translated into natural language prompts? How are LLM outputs validated as "coherent strategies"? What prevents hallucinated or inconsistent responses? How reproducible is this given LLM non-determinism?

[Addressed: a reproducibility-protocol paragraph is added at the end of this subsection answering all five questions (default model and local-model option, JSON state serialization, bounded-schema compilation with clamping, rule-based fallback on any invalid output, and the determinism policy).]

Beyond the built-in behavioral rules, an individual country’s policy may be set by a large-language-model agent. The agent receives the observed state of its country and its environment and returns a coherent strategy — domestic, foreign, financial, military, and trade policy — that is compiled into the model’s formalized actions. Several background-policy modes are supported: rule, growth target, language model, and a precompiled doctrine; the user sets the depth of strategy updating (per event, periodically, or once). A methodological distinction is important here: the generative model produces only the actors’ intentions and strategy, whereas their consequences are computed by the deterministic, empirically anchored core. The language model is responsible for the plausibility of behavior, the simulation core for the plausibility of the mechanics and numerical outcomes; this separation preserves reproducibility and shields the quantitative results from arbitrary judgments of the language model.

The protocol is specified for reproducibility. The default backend is an inexpensive hosted model (DeepSeek-chat at temperature 0.4); any OpenAI-compatible endpoint or a fully local model served through Ollama can be substituted, which removes the dependency on an external provider. The country’s observed state — its own economic, social, and military variables plus its bilateral relations — is serialized to a JSON payload and embedded in a fixed prompt template together with an explicit action schema listing every permissible field and its numeric bounds. The reply is parsed as JSON and compiled into the model’s typed policy actions; every numeric field is clamped to its schema bounds, and if parsing or validation fails at any point the agent falls back to the built-in rule-based policy for that turn — an invalid or hallucinated action can therefore never enter the simulation, it can only forfeit the language model’s influence for a step. Transport failures are retried; content failures are not, by design. Because provider-side sampling is not bit-reproducible, language-model agents are *off* in every validated configuration (the deterministic rule-based policy is the default), all headline numbers in this paper involve no language model anywhere in the loop, and multi-agent episodes archive the full prompt/response log so that any run can be audited and its consequences replayed exactly through the deterministic core.

Hybrid games and “what-if” hypotheses. The environment allows joint play by humans and agents: some countries are led by users, some by language agents or built-in rules, and intentions are stated in natural language and translated into model actions. Two move modes are provided. In *action* mode a move is committed and carried into the subsequent trajectory; in *“what-if”* mode a hypothesis is

played out in isolation, without irreversible consequences for the main line, which is convenient for testing counterfactual assumptions (for example, “how would the situation change if country X tightened sanctions against country Y”). Each move unfolds not into a single trajectory but into an ensemble, so that the response is measured together with its uncertainty — as in the scenario analysis of Section 7.

Interactive analytical environment. The graphical shell is a native desktop application over the same deterministic engine (frozen into a self-contained, offline binary): a natural-language assistant front end that maps a described scenario to grounded levers and returns per-country outcomes on a map; an expert mode with direct control over scenarios, ensembles, dose–response sweeps, sensitivity screening, and weak-signal scans; a run-comparison view against the validated baseline; and a live validation panel drawing the retrospective-backtest and conflict-ranking figures from the running engine. Every quantitative result comes from the engine, never from a language model; each chart displays the equivalent command-line invocation. The results of each run are exported in a readable form — a summary dashboard, an analytical brief for decision-makers, and logs of policies and baseline trajectories — which makes a multi-agent episode traceable and amenable to subsequent review.

The scenario environment thus operationalizes the model’s stated purpose. The coupled core turns from a computational mechanism into a platform for exercising multi-agent interaction: a collective of language agents or a mixed human–agent composition plays out specific scenarios and counterfactual hypotheses, while the user observes the propagation of shocks through the economy, climate, society, and geopolitics and the early weak signals of approaching structural shifts — within a single, reproducible, auditable loop. The role of generative AI here is auxiliary but substantive: it expands the space of plausible actor strategies without supplanting the empirically anchored mechanics of consequences.

9. A game-mechanical simulation on the model core

The scenario environment of Section 8 has been taken one step further: the model core has been wrapped into a complete decision game (working title *GODMODE*), in which a single player steers the world from 2023 toward 2050 through roughly thirty annual turns. Each year a decision card poses a dilemma drawn from a deck of 158 situations (109 shocks, 49 opportunities); the chosen option is compiled into the same typed policy actions available to the model’s agents (fiscal, monetary, energy, trade, military, and research levers), the full 57-actor engine steps one year, and the consequences the player sees — output, prices, tension, trust, crises, wars — are the engine’s, not scripted outcomes. A run is lost the moment any of seven catastrophe thresholds is crossed (temperature, biosphere health, social stability, economic and geopolitical strain, energy and food prices); reaching 2050 with all gates intact is a win. The game layer adds only presentation and pacing: every quantitative consequence is computed by the deterministic, validated core, one engine year costing 30–50 ms, so a full game simulates in about a second — which is what makes large-scale balance testing statistically meaningful.

We report the game here for two reasons: its outcome distributions are themselves a stress test of the coupled model under sustained, adversarially chosen policy sequences; and balancing it uncovered structural properties of the model — including two genuine defects — that neither the retrospective backtests nor the ensembles had surfaced.

Outcome distributions. The balance instrument is batches of automated playthroughs by scripted policy archetypes: *green* (always the more decarbonising option), *growth* (always the more expansionary), *mixed* (uniformly random choices), and *passive* (always the first option offered). In the reference batch of 1,002 games (334 per archetype, deterministic seeds; Fig. 6) the overall survival rate is 23.5%, within the intended difficulty band of roughly one win in four to six attempts, and losses are spread across causes — 35% climate, 22% economic strain, 21% geopolitical strain, 13% social stability, 8% energy price — with no single dominant failure mode, itself a tuned property (early builds funnelled every loss through one channel). Failures cluster near the end of the run: 77% of losing games fail in 2040–2045, within sight of the finish, because the quantities that kill are slow accumulators (temperature, debt, tension) that a decade of comfortable early play quietly loads.

Emergent regularities. Four patterns proved robust across batches and are, we would argue, properties of the underlying model rather than of the card deck. *First, no doctrinaire strategy dominates.* The mixed archetype survives most often (28.4%), beating both consistent green (20.1%) and consistent growth

(21.9%), and each doctrine dies of its characteristic disease: green runs are killed by economic strain (the transition cost of aggressive decarbonisation — 43 % of green losses), growth runs by climate and collapsing social stability (39 % and 34 % of theirs). Single-axis optimisation shifts the binding constraint to another subsystem — the game-mechanical face of the coupling this paper is about. *Second, passivity is the worst strategy.* The always-first-option archetype wins only 18.7 % of the time and dies overwhelmingly of economic strain (67 % of its losses): in a stock-flow-consistent world, drifting means accumulating debt. *Third, restraint on the strain axes dominates any single lever.* In a controlled 40-seed experiment, a policy prudent on both the economic and geopolitical strain axes at once wins about 75 % of runs, while one reckless on both (maximal military posture and deficits) wins about 5 % — a spread far wider than any single-lever effect, now pinned by a regression test. *Fourth, assistance mechanics exhibit a moral-hazard cost:* a “regional stabilization package” that buys an ailing region time measurably lowers the final win rate when the underlying driver is left unaddressed — borrowed stability rebounds.

The game as a diagnostic instrument. The most consequential product of balancing was not a tuning constant but two model fixes. Play sessions showed, first, that world resource prices drifted to their clamps and stayed there regardless of policy; tracing the cause exposed a missing equilibrium term in the price rule and two quantity-side accounting defects (a frozen food-demand base and a compounding recycling term), fixed in release v18.1.2 — after which prices became genuinely two-sided in play. Second, hawk-and-dove experiments showed government trust declining on statistically indistinguishable paths under opposite policies — a structural attractor caused by an inequality drag roughly forty times stronger than the only positive term in the trust equation; the fix (an opt-in equilibrium anchor for trust, v18.1.3) lifted the floor and removed a spurious post-collapse oscillation. Neither defect was visible in the 2015–2023 backtests — both live in corners of state space that only long, adversarial trajectories visit. Sustained interactive play explores exactly the regions that validation against smooth history does not, which is why we regard the game layer as part of the model’s quality programme, not an entertainment appendix. Difficulty is tuned against the same outcome batches through a threshold-flip analysis (by how much a candidate gate would reclassify past winners: tightening the warming gate to 1.8 °C flips 59 % of wins to losses, a 180-point food-price gate 45 %), so balance decisions are made on measured distributions rather than designer intuition. Human playtest groups have so far contributed qualitative interface-and-mechanics feedback rather than statistically usable outcome data — the distributions above are scripted-bot distributions, a stated limitation; collecting a clean human sample is ongoing work.

10. Conclusion

The model presented here links the economy, climate, natural capital, society, politics, geopolitics, and culture within a single loop and thereby fills a substantive gap between specialized single-domain models. Its value for strategic planning, scenario analysis, and risk management lies not in point forecasting but in the ability to play out scenarios coherently, compare alternative decisions, express uncertainty correctly through ensembles, and detect weak signals of approaching structural shifts. The economic and climate blocks are empirically anchored and pass retrospective validation; the socio-political and geopolitical blocks, which constitute the model’s distinguishing feature, are validated against a more modest but substantive criterion — improvement over naive benchmarks. The economic loop has since been deepened: an explicit money-to-prices transmission (a calibrated quantity-theory term), an endogenous long-run-growth mechanism (R&D-stock accumulation à la Jones), a tested forward-looking-expectations mechanism (left off by default, as its marginal effect proved unstable), and a data-grounded military-capability index (SIPRI military expenditure as a CINC-style component, validated against the post-2022 militarization record) have all been added. A game-mechanical layer over the same core (Section 9) has proved to be more than an interface: its outcome distributions behave as a stress test of the coupled dynamics (no doctrinaire strategy dominates; each dies of its characteristic cause), and balancing it surfaced — and led to the repair of — two structural degeneracies invisible to retrospective validation. The residual limitations — the fast-approximation climate block, the discipline-wide uncertainty of damage estimates, and the intrinsically low predictability of conflict — are stated honestly and are not obstacles to applying the model in the modes outlined.

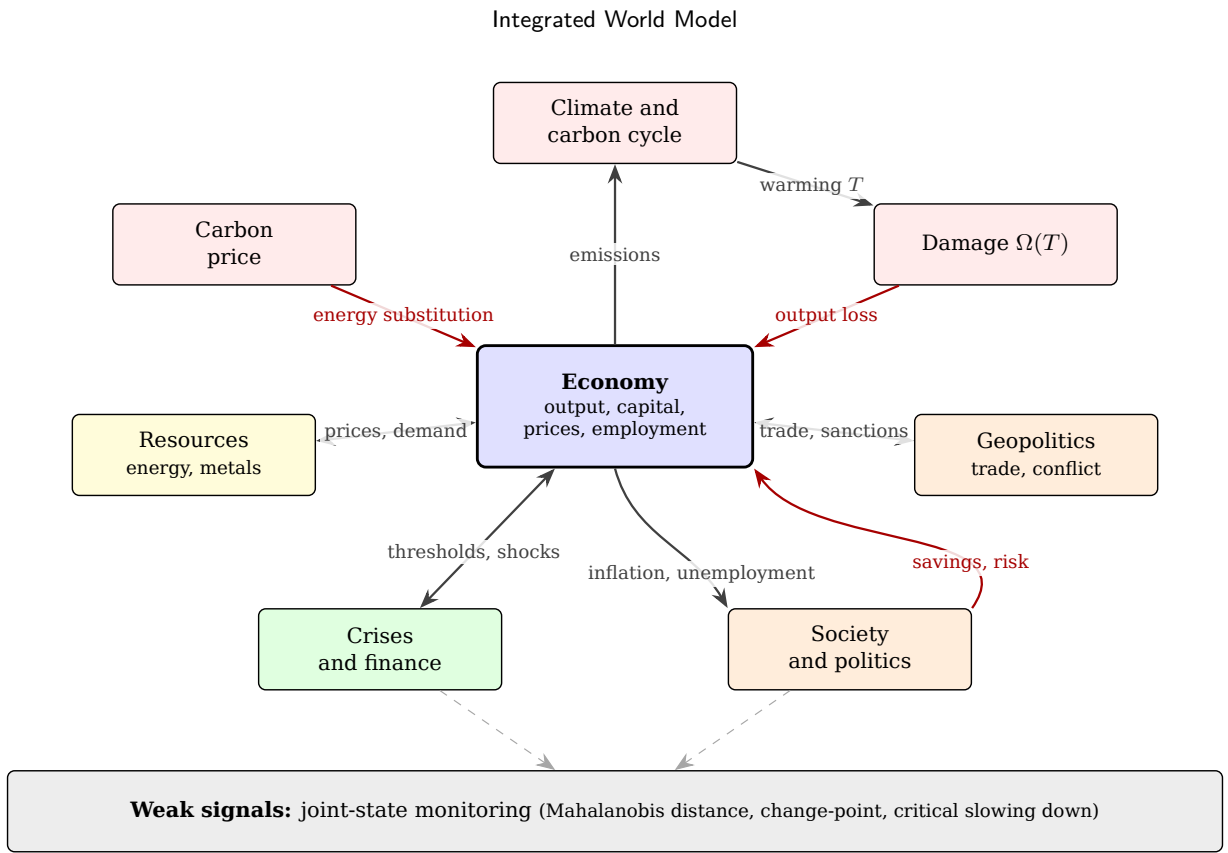


Figure 1: Model architecture and the links between blocks. The economy is the central hub; black arrows show direct effects, red arrows show feedbacks that reduce output or savings. The closed climate loop (from emissions to warming, then to damage and reduced output) is complemented by the carbon-price channel, the resource, socio-political, and geopolitical blocks, and the discrete-crisis block. The bottom layer implements continuous monitoring of the joint state to detect weak signals.

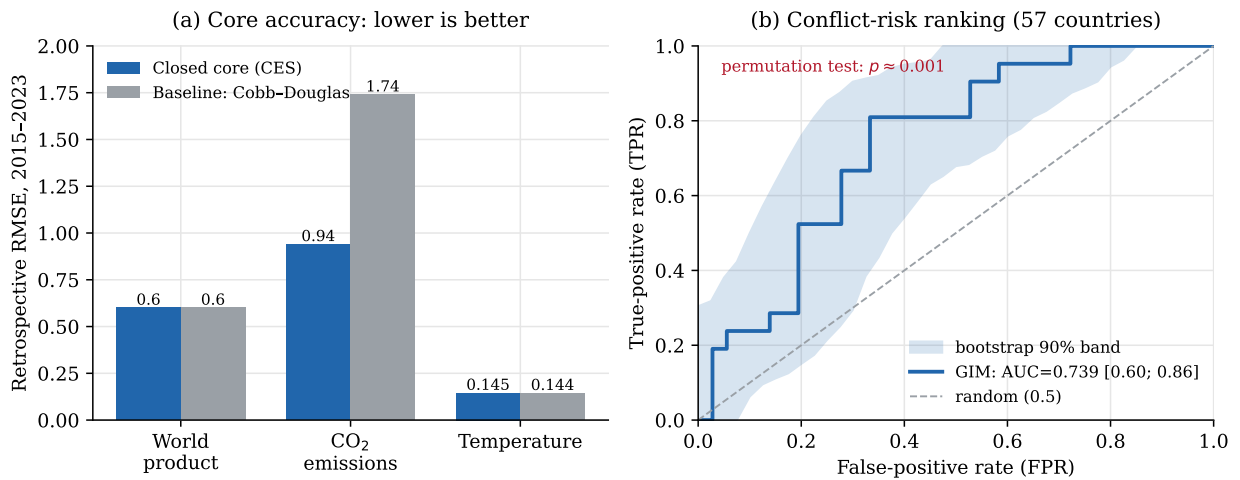


Figure 2: Retrospective-validation results. (a) Root-mean-square error of the 2015–2023 retrospective check for the closed core (nested CES) compared with the former Cobb–Douglas core, for world product, CO₂ emissions, and temperature (lower is better). (b) ROC curve of the conflict block: ranking 57 countries by conflict-proneness against the registry of armed conflicts of 1990–2023 yields an area under the curve of 0.739 (the bootstrap 90 % band is shaded; 95 % CI [0.60; 0.86]; permutation test $p \approx 0.001$) versus 0.5 for random guessing.

Sensitivity screening (Morris, 33 priors): seven leading parameters

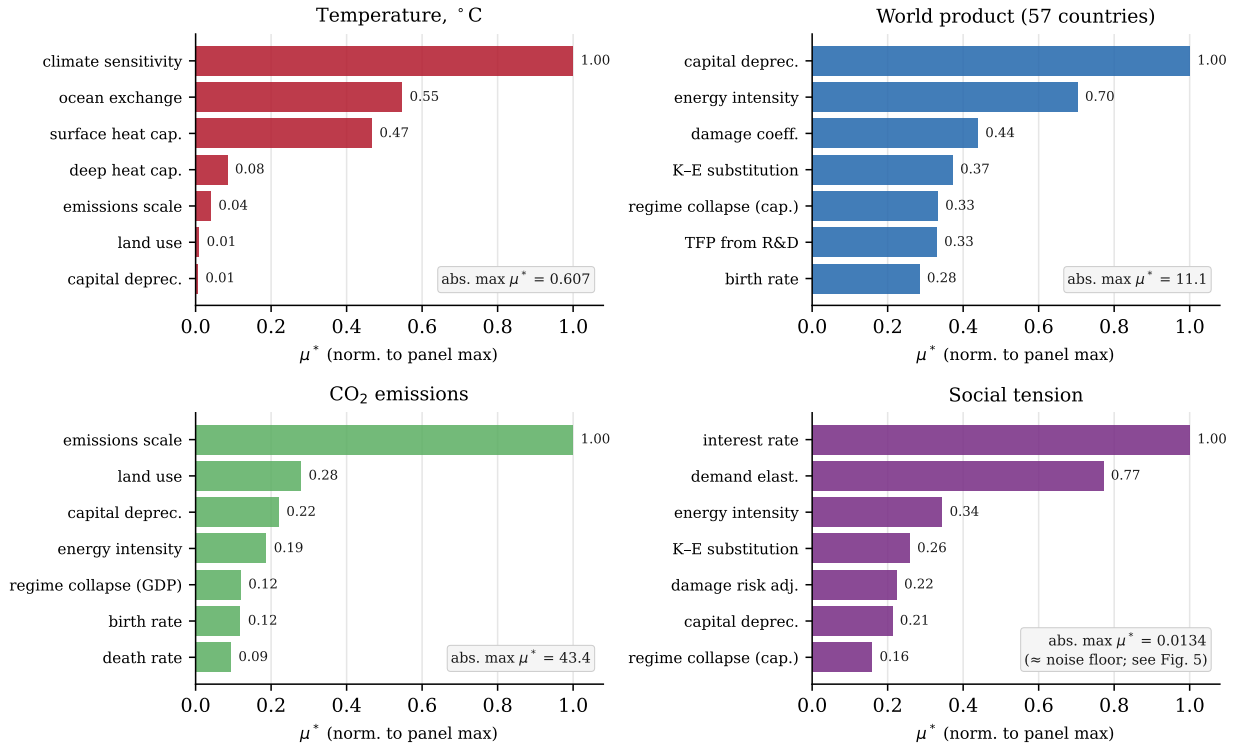


Figure 3: Global sensitivity screening (Morris method) over 33 key parameters for four output quantities (57 countries); the seven most influential parameters are shown, with μ^* normalized to the maximum within each panel and the absolute maximum μ^* indicated in the box. Temperature is governed by climate parameters, product and emissions by economic and emission parameters. For social tension the absolute μ^* is about three orders of magnitude lower (at the noise floor): it is almost insensitive to the physico-economic priors, and its own channel is examined in Fig. 5.

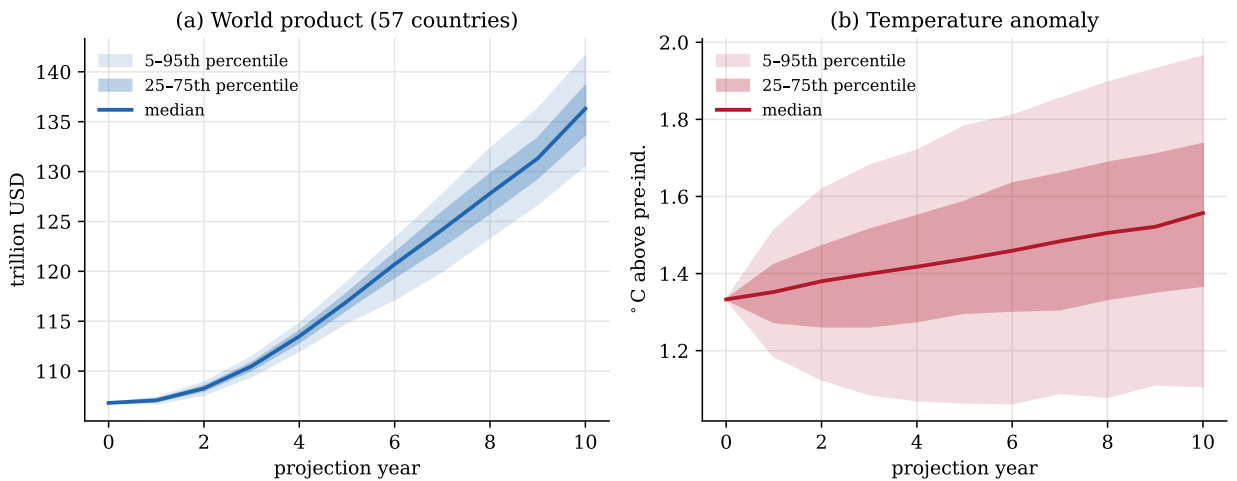


Figure 4: Ensemble uncertainty quantification (500 members): the distribution of outcomes from a Monte Carlo run with random parameter draws from the prior ranges. The median and the 25–75th- and 5–95th-percentile bands are shown for (a) the aggregate product of the 57 modeled countries and (b) the temperature anomaly.

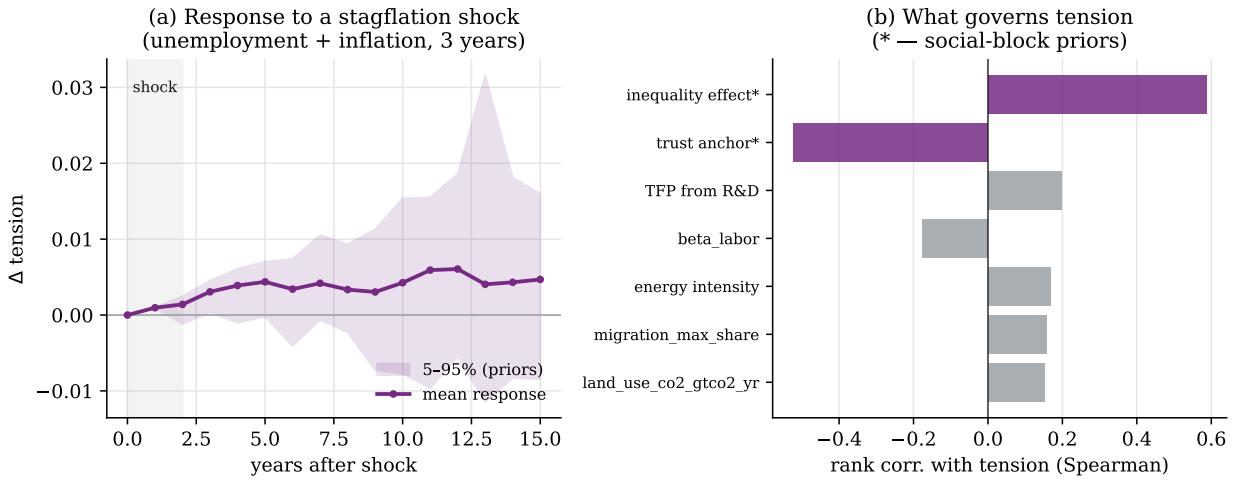


Figure 5: The influence of economic conditions on social tension. (a) Impulse response of mean tension to an exogenous stagflation shock (unemployment and inflation, three years): the mean curve and the 5–95th-percentile band over priors; the response builds slowly and persists beyond a decade. (b) Localization: the rank correlation of priors with final tension; the parameters of the social block (marked *) dominate, not the physico-economic ones.

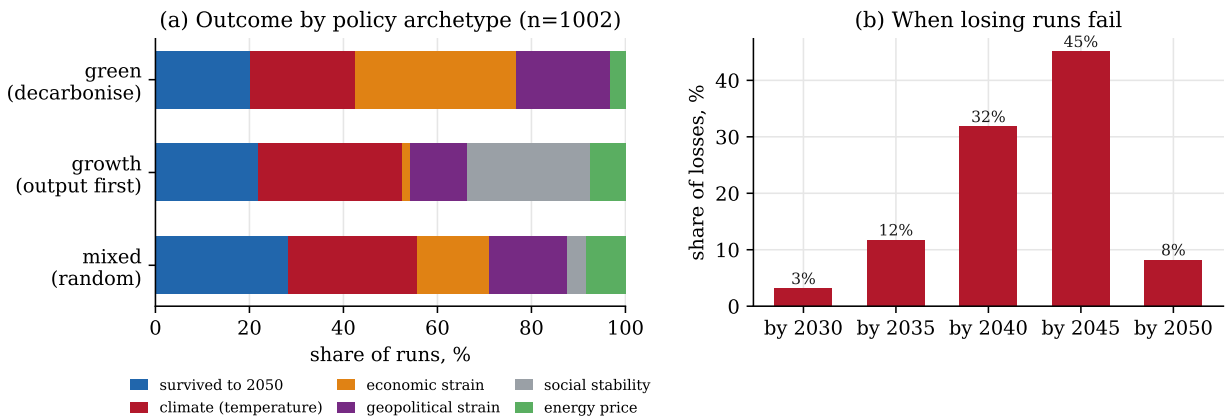


Figure 6: Game-mechanical simulation on the model core (1,002 automated playthroughs, 2023–2050). (a) Outcome composition by scripted policy archetype: survival share and loss causes. No doctrinaire strategy dominates, and each dies of its characteristic cause (green of economic strain, growth of climate and social stability). (b) Timing of failures: 77% of losing runs fail in 2040–2045, as slow accumulators (temperature, debt, tension) come due.

Parameter	Central	Range	Source
Equilibrium climate sensitivity, °C	3.0	1.5–6.0	IPCC AR6 WG1 (2021); Otto et al. 2013
Surface heat capacity	8.0	6.0–18.0	Geoffroy et al. 2013 (two-layer EBM)
Ocean heat exchange	1.0	0.4–1.3	Geoffroy et al. 2013
Emissions scale	0.98	0.90–1.05	Global Carbon Project 2023
Structural decarbonization rate	0.016	0.010–0.030	GCP / WDI carbon-intensity trend 2000–2023; GIM 2015–2023 backtest
Capital share α	0.30	0.22–0.40	Penn World Table 10; Gollin 2002
Labor share β	0.60	0.50–0.72	Penn World Table 10; Gollin 2002
Energy elasticity γ	0.042	0.015–0.080	2015–2023 backtest; energy literature
Capital–energy substitution σ_{KE}	0.40	0.20–0.60	van der Werf 2008; Koetse et al. 2008
Depreciation rate δ	0.05	0.03–0.08	Penn World Table 8/9
TFP sensitivity to R&D	0.30	0.10–0.60	Calibration to backtest
Damage coefficient	0.0078	0.0015–0.025	Howard & Sterner 2017 (productivity-corrected central); DICE-2016R2; Burke et al. 2015
Pure rate of time preference ρ	0.015	0.001–0.030	DICE-2016R2; Stern 2007
Elasticity of marginal utility η	1.45	1.0–2.0	DICE-2016R2; Stern 2007
Market demand elasticity	0.40	0.20–0.70	Labandeira et al. 2017 (meta-analysis)
<i>Social block (structural assumptions; validated by sign and channel, Section 5)</i>			
Tension stress from unemployment	0.010	0.007–0.014	Structural assumption (higher unemployment raises discontent)
Tension stress from inflation	0.005	0.003–0.007	Structural assumption
Inequality effect on tension	0.0005	0.0003–0.0007	Structural assumption
Trust-anchor strength	0.060	0.039–0.081	Structural assumption (reversion to equilibrium)

A. Key parameters and their identification

Each parameter is specified by a prior distribution; below are the central value, the plausibility range (roughly the 5–95th percentile of the prior), and the main source. The 33 key parameters carry literature-grounded priors (each with a source and rationale); the model’s full registry comprises 287 parameters with calibration-status tags, most of them expert-set or structural, and is available in the repository (Section 8).

B. Integration benchmark: where sectoral models stop

The breadth-of-coupling claim (Table 1) is made *computational* here. The design is literature-anchored rather than an internal ablation: for each of three shocks we take a published conclusion from a named sectoral model, feed GIM the *same* exogenous shock, and ask whether GIM (a) reproduces the sectoral first-order effect and (b) surfaces a cross-sector consequence the sectoral model cannot represent by construction. The unit of comparison is the *decision-relevant conclusion*, not a headline number: GIM is not claimed to compute emissions or welfare “better” than DICE, but to see a consequence DICE is structurally blind to. Runs are deterministic (the full 57-country world, a ten-year horizon, extreme events off, so the only difference between the baseline and shock arms is the injected shock); each scenario also reports a dose–response sweep on which the sectoral model’s slope is, by construction, zero. The harness and figures are reproducible from the repository (Section 8).

Two methodological points are stated plainly, as they are part of the result. First, GIM’s aggregate resource-*price* subsystem saturates within one simulated year, so a purely price-mediated shock does not propagate forward; each shock is therefore injected at its first cross-sector transmission variable — precisely the channel a sectoral model lacks (a consumer-price wedge for the carbon tax, an import-bill and reserve burden for the oil shock, a food-supply gap for the crop shock). Second, the three spillovers differ in magnitude and shape, and are reported as such rather than uniformly inflated.

B.1. Carbon tax (anchor: DICE)

DICE-2016R [Nordhaus, 2017] finds a welfare-optimal carbon price (about \$31/tCO₂ in 2015, rising) whose only first-order effect is lower emissions; it contains no labour market and no political economy. Fed the same \$50/tCO₂ tax, GIM reproduces the emissions decline (about 5.5% over ten years) and, in addition, raises social tension and lowers government trust as the tax’s pass-through to energy prices feeds inflation and unemployment — the political-economy channel behind the reversals of the French carbon and fuel tax (2018) and the Australian carbon price (2014). As the rate increases, the tension response rises monotonically (Fig. 7), while DICE’s response on that axis is zero.

B.2. Oil price shock (anchor: energy-system models)

An energy-system model such as MESSAGEix resolves an oil-supply shock as a price rise and demand adjustment, bounded within the energy sector. GIM carries the same shock further, into sovereign finance: the higher import bill, financed by drawing reserves and issuing debt, brings the most fiscally fragile importers to their debt-crisis thresholds, and which countries tip is decided by the model’s own debt and interest-rate triggers, not by hand. At a realistic shock (a 20% supply cut, a burden of about 3% of GDP) the effect is marginal: two additional importers enter crisis. Under a severe shock (a burden of about 15% of GDP, comparable to the worst historical oil episodes) the mean importer debt-crisis exposure rises about threefold and four further countries enter sovereign crisis (Fig. 8). The response is threshold-stepped, and the energy-system model produces no sovereign crises at any magnitude. This is the mechanism behind episodes such as Sri Lanka’s 2022 default.

B.3. Crop yield shock (anchor: AgMIP and AR6)

Crop assessments in the AgMIP project and the IPCC Sixth Assessment stop at yield decline and a food-security index. GIM carries the same yield loss further, into domestic instability: cutting the production and reserves of the most income-vulnerable importers opens a food-supply gap that raises protest pressure, and the escalation accelerates as the loss deepens. This is the nonlinear “crossing into high impact” documented for food prices and the Arab Spring [Lagi et al., 2011] (a threshold FAO food-price index of 210). At a severe 50% regional yield loss, food-affordability stress rises by about 0.11 and protest pressure by about 0.016 on the vulnerable subset (Fig. 9); further along the chain the response attenuates toward regime fragility, and the endpoint is domestic instability rather than interstate conflict.

Across the three channels (Fig. 10) GIM exhibits a positive, measurable slope on an axis where every sectoral model is, by construction, flat at zero. That gap — not any single number — is the integration dividend, and it is exactly the dimension the coverage table summarizes qualitatively.

Scenario A — Carbon tax \$50/tCO₂: GIM agrees on emissions, adds the political-economy cost

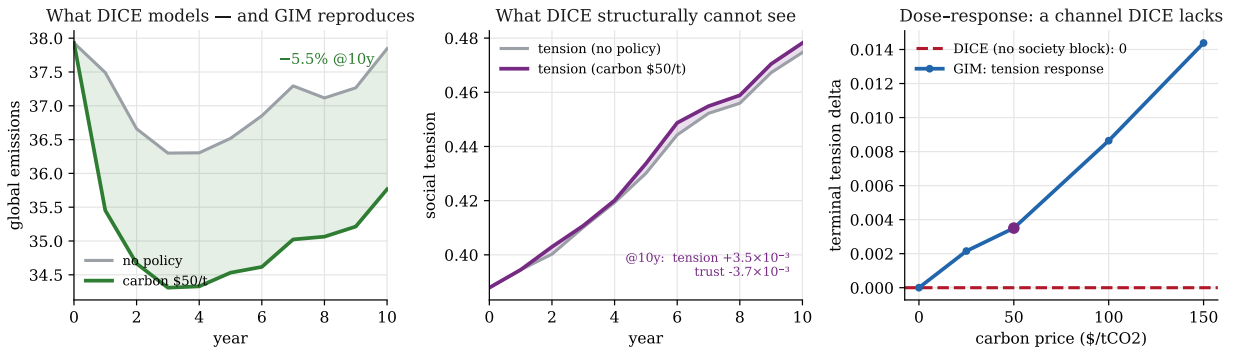


Figure 7: Carbon-tax scenario. Left: GIM reproduces the emissions decline that DICE models. Middle: GIM adds the social-tension response DICE cannot represent. Right: the response of tension to the tax rate has a non-zero GIM slope against DICE's zero response.

Scenario B — Oil price shock: an energy model sees price; GIM sees the sovereign-debt cascade

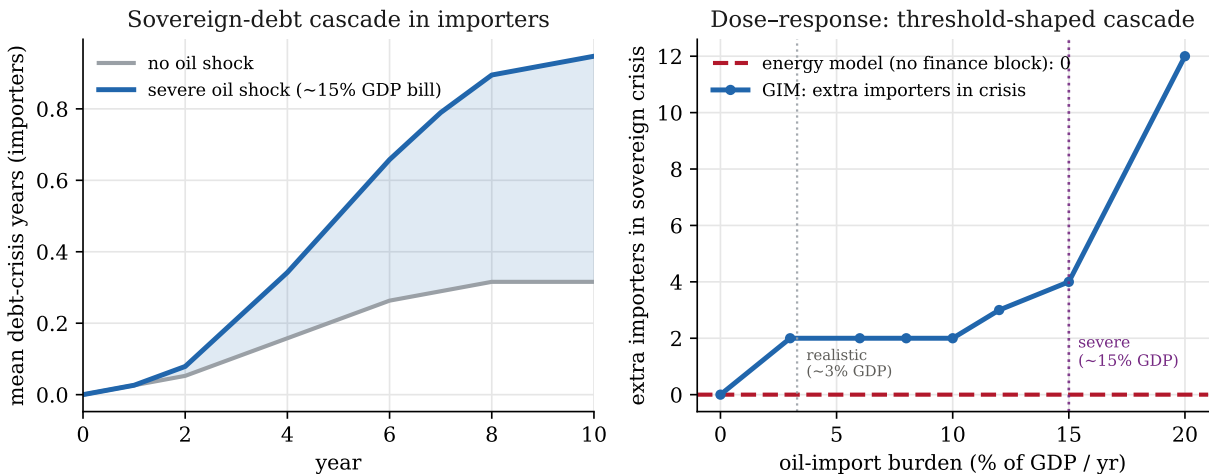


Figure 8: Oil-price-shock scenario. Left: importer sovereign-debt-crisis duration under a severe shock. Right: the response to the import burden is threshold-stepped, marking the marginal (realistic) and pronounced (severe) points against the energy-system model's zero response.

Scenario C — severe crop yield shock: AgMIP stops at hunger; GIM carries it to protest pressure

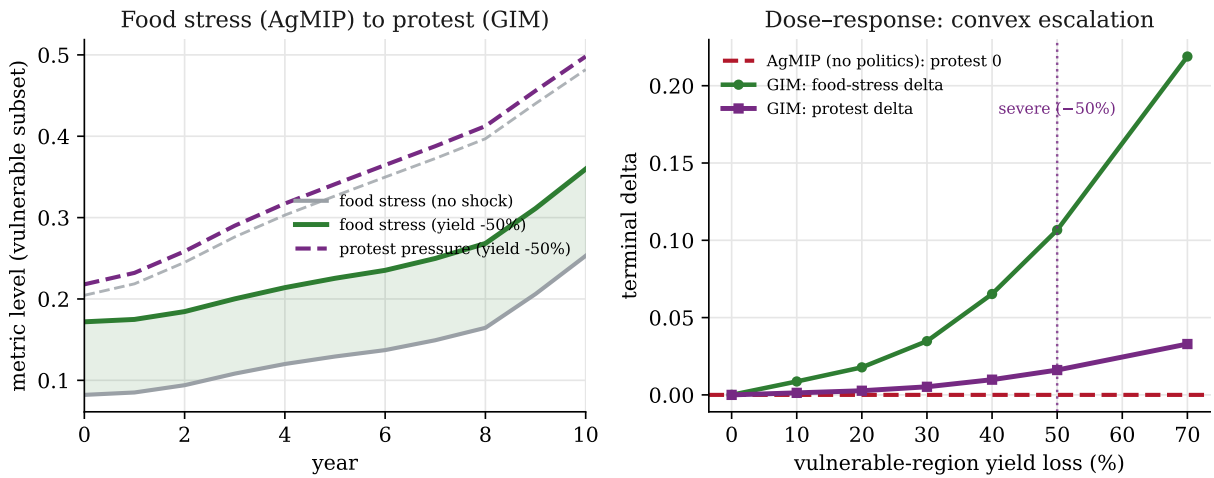


Figure 9: Crop-yield-shock scenario. Left: food-affordability stress and protest pressure on the vulnerable subset. Right: the response to the depth of the yield loss is convex against AgMIP's zero response on the protest axis.

The integration dividend — three cross-sector channels with a non-zero GIM slope where sectoral models are flat

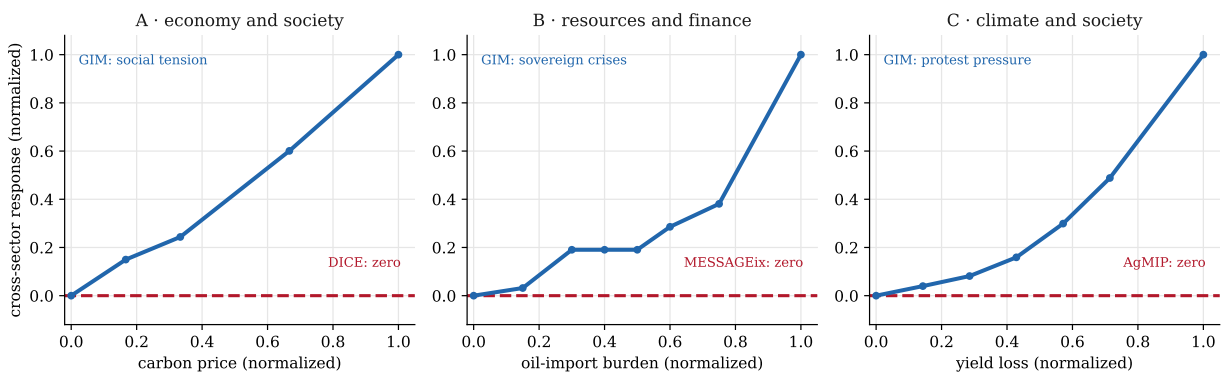


Figure 10: The integration dividend. Each panel is one cross-sector channel, normalized to its own maximum; the dashed line is every sectoral model's zero response on that axis. GIM's slope is smooth for the economy and society channel, threshold-stepped for the resources and finance channel, and convex for the climate and society channel.

References

- Vivek Arora and Martin Cerisola. How does U.S. monetary policy influence sovereign spreads in emerging markets? *IMF Staff Papers*, 48(3):474–498, 2001.
- Kenneth J. Arrow, Hollis B. Chenery, Bagicha S. Minhas, and Robert M. Solow. Capital-labor substitution and economic efficiency. *The Review of Economics and Statistics*, 43(3):225–250, 1961. doi: 10.2307/1927286.
- Ben S. Bernanke, Mark Gertler, and Simon Gilchrist. The financial accelerator in a quantitative business cycle framework. In John B. Taylor and Michael Woodford, editors, *Handbook of Macroeconomics*, volume 1, pages 1341–1393. Elsevier, 1999.
- Marshall Burke, Solomon M. Hsiang, and Edward Miguel. Global non-linear effect of temperature on economic production. *Nature*, 527:235–239, 2015. doi: 10.1038/nature15725.
- Marshall Burke, Mustafa Zahid, Noah S. Diffenbaugh, and Solomon Hsiang. Quantifying climate change loss and damage consistent with a social cost of greenhouse gases. NBER Working Paper 31658, National Bureau of Economic Research, 2023.
- Francesca Campolongo, Jessica Cariboni, and Andrea Saltelli. An effective screening design for sensitivity analysis of large models. *Environmental Modelling & Software*, 22(10):1509–1518, 2007. doi: 10.1016/j.envsoft.2006.10.004.
- Oya Celasun et al. Forecasting the imf's world economic outlook: An evaluation. *IMF Working Paper*, 2021.
- Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, New York, 1993.
- Pierre Friedlingstein et al. Global carbon budget 2023. *Earth System Science Data*, 15(12):5301–5369, 2023. doi: 10.5194/essd-15-5301-2023.
- Olivier Geoffroy, David Saint-Martin, Dirk J. L. Olivié, Aurore Voldoire, Gilles Bellon, and Sophie Tytéca. Transient climate response in a two-layer energy-balance model. part I: Analytical solution and parameter calibration using CMIP5 experiments. *Journal of Climate*, 26(6):1841–1857, 2013. doi: 10.1175/JCLI-D-12-00195.1.
- Wynne Godley and Marc Lavoie. *Monetary Economics: An Integrated Approach to Credit, Money, Income, Production and Wealth*. Palgrave Macmillan, Basingstoke, 2007.
- Keith Head and Thierry Mayer. Gravity equations: Workhorse, toolkit, and cookbook. In *Handbook of International Economics*, volume 4, pages 131–195. Elsevier, 2014. doi: 10.1016/B978-0-444-54314-1.00003-3.
- Jens Hilscher and Yves Nosbusch. Determinants of sovereign risk: Macroeconomic fundamentals and the pricing of sovereign debt. *Review of Finance*, 14(2):235–262, 2010. doi: 10.1093/rof/rfq005.
- Geert Hofstede, Gert Jan Hofstede, and Michael Minkov. *Cultures and Organizations: Software of the Mind*. McGraw-Hill, New York, 3 edition, 2010.
- Peter H. Howard and Thomas Sterner. Few and not so far between: A meta-analysis of climate damage estimates. *Environmental and Resource Economics*, 68:197–225, 2017. doi: 10.1007/s10640-017-0166-z.
- Solomon M. Hsiang, Marshall Burke, and Edward Miguel. Quantifying the influence of climate on human conflict. *Science*, 341(6151):1235367, 2013. doi: 10.1126/science.1235367.
- IPCC. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, 2021.
- Charles I. Jones. R&d-based models of economic growth. *Journal of Political Economy*, 103(4):759–784, 1995. doi: 10.1086/262002.
- Fortunat Joos, Raphael Roth, Jan S. Fuglestad, et al. Carbon dioxide and climate impulse response functions for the computation of greenhouse gas metrics: a multi-model analysis. *Atmospheric Chemistry and Physics*, 13:2793–2825, 2013. doi: 10.5194/acp-13-2793-2013.
- Mark J. Koetse, Henri L. F. de Groot, and Raymond J. G. M. Florax. Capital-energy substitution and shifts in factor demand: A meta-analysis. *Energy Economics*, 30(5):2236–2251, 2008. doi: 10.1016/j.eneco.2007.06.006.
- Marco Lagi, Karla Z. Bertrand, and Yaneer Bar-Yam. The food crises and political instability in north africa and the middle east. *arXiv preprint arXiv:1108.2455*, 2011.
- Prasanta Chandra Mahalanobis. On the generalised distance in statistics. *Proceedings of the National Institute of Sciences of India*, 2(1):49–55, 1936.
- Donella H. Meadows, Dennis L. Meadows, Jørgen Randers, and William W. Behrens. *The Limits to Growth*. Universe Books, New York, 1972.
- Malte Meinshausen, Sarah C. B. Raper, and Tom M. L. Wigley. Emulating coupled atmosphere-ocean and carbon cycle models with a simpler model, MAGICC6. *Atmospheric Chemistry and Physics*, 11:1417–1456, 2011. doi: 10.5194/acp-11-1417-2011.
- Jean-Francois Mercure, Hector Pollitt, Jorge E. Vinuales, Neil R. Edwards, Philip B. Holden, Unnada Chewprecha, Pablo Salas, Ida Sognnaes, Aileen Lam, and Florian Knobloch. Macroeconomic impact of stranded fossil fuel assets. *Nature Climate Change*, 8:588–593, 2018. doi: 10.1038/s41558-018-0182-1.
- Millennium Institute. Threshold 21 (T21) overview. Technical report, Millennium Institute, Arlington, VA, 2002.
- Ashoka Mody, Franziska Ohnsorge, and Damiano Sandri. Precautionary savings in the great recession. *IMF Economic Review*, 60(1):114–138, 2012. doi: 10.1057/imfer.2012.5.
- Max D. Morris. Factorial sampling plans for preliminary computational experiments. *Technometrics*, 33(2):161–174, 1991. doi: 10.1080/00401706.1991.10484804.
- William D. Nordhaus. Revisiting the social cost of carbon. *Proceedings of the National Academy of Sciences*, 114(7):1518–1523, 2017. doi: 10.1073/pnas.1609244114.
- Kevin Rennert et al. Comprehensive evidence implies a higher social cost of CO₂. *Nature*, 610:687–692, 2022. doi: 10.1038/s41586-022-05224-9.

- Keywan Riahi et al. The shared socioeconomic pathways and their energy, land use, and greenhouse gas emissions implications: An overview. *Global Environmental Change*, 42:153–168, 2017. doi: 10.1016/j.gloenvcha.2016.05.009.
- Marten Scheffer et al. Early-warning signals for critical transitions. *Nature*, 461:53–59, 2009. doi: 10.1038/nature08227.
- Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978. doi: 10.1214/aos/1176344136.
- J. David Singer, Stuart Bremer, and John Stuckey. Capability distribution, uncertainty, and major power war, 1820–1965. *Peace, War, and Numbers*, pages 19–48, 1972.
- Christopher J. Smith, Piers M. Forster, Myles Allen, Nicholas Leach, Richard J. Millar, Giovanni A. Passerello, and Leighton A. Regayre. FAIR v1.3: a simple emissions-based impulse response and carbon cycle model. *Geoscientific Model Development*, 11:2273–2297, 2018. doi: 10.5194/gmd-11-2273-2018.
- Ralph Sundberg and Erik Melander. Introducing the UCDP georeferenced event dataset. *Journal of Peace Research*, 50(4): 523–532, 2013. doi: 10.1177/0022343313484347.
- Nassim Nicholas Taleb. *The Black Swan: The Impact of the Highly Improbable*. Random House, New York, 2007.
- John B. Taylor. Discretion versus policy rules in practice. *Carnegie-Rochester Conference Series on Public Policy*, 39:195–214, 1993. doi: 10.1016/0167-2231(93)90009-L.
- U.S. Environmental Protection Agency. Report on the social cost of greenhouse gases: Estimates incorporating recent scientific advances. Technical report, U.S. Environmental Protection Agency, 2023.
- Edwin van der Werf. Production functions for climate policy modeling: An empirical analysis. *Energy Economics*, 30(6): 2964–2979, 2008. doi: 10.1016/j.eneco.2008.05.008.
- Martin L. Weitzman. On modeling and interpreting the economics of catastrophic climate change. *The Review of Economics and Statistics*, 91(1):1–19, 2009. doi: 10.1162/rest.91.1.1.
- Daniel Williamson et al. History matching for exploring and reducing climate model parameter space using observations and a large perturbed physics ensemble. *Climate Dynamics*, 41:1703–1729, 2013. doi: 10.1007/s00382-013-1896-4.
- Edwin B. Wilson and Margaret M. Hilferty. The distribution of chi-square. *Proceedings of the National Academy of Sciences*, 17(12):684–688, 1931. doi: 10.1073/pnas.17.12.684.